



Debre Tabor University

Faculty of Technology

Department of computer science

Operating System Laboratory Manual

For

Third Year Computer Science Students

Compiled by:

Huluager W. and Genet W.

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Introduction to DOS Commands

The Disk Operating System (DOS) allows the user to communicate with the computer. When you “Boot” the computer you load DOS, which in turn consists of a number of components, all of which enable the user to perform the various functions, such as printing, listing directories, managing files, etc. One of the major components DOS loads is called **command.com** of which one function is to load into memory a collection of the most often used DOS commands. For example, one of these commands is “DIR” which displays the listing of files in the default (presently active) directory. If, however, all DOS commands would be incorporated into command.com it would be too large and take up too much memory. Consequently, no other programs (eg. Word Processor, Games, etc.) Could be used. For this reason, DOS separated a number of programs (commands) into individual units. These are referred to as “External Commands” as opposed to those residing in command.com which are called “Internal Commands”.

The External commands are the commands that are available only from the disk where they are located and can only be activated by directing the computer to their location and then typing their name. The Internal commands can be activated at any time simply by typing their name.

All this information is not important for the school computers because they have been setup using the PATH command (more on the PATH command later) which automatically directs the computer to the location where the External DOS commands are stored. This information is included here, however, for home use. If you use these commands on your computer at home and the commands are not working properly then the reason is probably that you are using an External command and you have not properly directed the computer to the location where these commands are stored.

Format

Before using a disk, either a floppy disk or a fixed disk, the disk must be formatted.

Formatting the disk lays down a series of segments and tracks that the computer uses to locate files saved on the disk. If you have a disk which already contains files, formatting the disk will erase these files. As a minimum the format command must tell the location of the disk to be formatted. For example, `format a:` will format the disk in drive a:

Note be very careful when formatting a disk. You must direct the format to the proper drive. In the example to the left, if you mistaking type `format c:` instead of `format a:` you will format the disk at c: which is your hard drive and thus you will lose all files stored there.

Switches

Switches allow the user to select extra options when working with MS-DOS commands. The option tells the computer to do extra work or to handle things in a certain manner. Switches vary from command to command.

Syntax: Format [drive:] /s installs the DOS system files on your disk so the disk can be used to boot the computer. It should be noted that this command may not work on the network. If it does not work, simply format the disk with the /v switch.

Format [drive:] /v allows the user to give the disk a name.

Dir

The Directory command allows the user to see the index of the disk. This permits the user to verify if there are files on the disk and if so what these files are. We will deal with the Directory command in more detail in later lessons but for the moment it is handy to be able to see the index of the disk in order to see the name of our disk and the files which have been copied as a result of the /s switch if the /s switch worked.

Syntax: Dir [drive:] shows the index for the disk in the specified drive.

Changing drives

When first started, the computer uses the drive on which it found DOS as the current drive. This may not be the drive on which you would like to work. To change drives, type the drive letter on which you would like to work followed by a colon, and press ENTER.

Syntax: c:\>[drive:] Changes from the c: drive to the drive specified.

a:\>[drive:] Changes from the a: drive to the drive specified.

b:\>[drive:] Changes from the b: drive to the drive specified

At the c:\> prompt type D: and press ENTER

At the a:\> prompt type C:

Label

The Label command is used to put a label identifier on a disk. After formatting a disk a prompt will request you to place a label on a disk. The label may be entered, a person's name is often used, or the ENTER key pressed if no label is needed. In order to re-label or delete a label one need not reformat the disk. By

using the Label command the "name" on the disk can be changed or deleted.

Syntax: Label allows the user to label the disk on the current drive.

Label [drive:] allows the user to label the disk in the specified drive.

Vol

The **vol** command is used to check to see what label has been placed on a disk.

Syntax: Vol Allows the user to see the name of the disk in the current drive.

Vol [drive:] Allows the user to see the name of the disk in the specified drive.

Ver

The **ver** command is used to determine which version of MS-DOS is being run in the computer. This is important to know as some DOS commands have been replaced in later versions of DOS, for example **edlin**, and new commands have been introduced in later versions. The command **xcopy** is not available in some early DOS versions.

Syntax: Ver displays the version of Dos that is currently running.

CLS

The **cls** command is used to clear the screen. This command is helpful as it will remove any distracting information that is present on the screen.

Syntax: Cls Clears the screen.

Date and Time

The date and time commands can be used to set the clock in the computer. The clock in a computer is not precise; it will not keep time accurately. You may also want to change the time in your computer when the clocks change for daylight saving time. You may also need to change the date in your computer.

Syntax: Time allows the user to see the time and to make changes

Type date and press ENTER. Notice that the computer gives a time and then asks you to enter a time. If the time is correct press ENTER. If the time is not correct, enter the correct time in the format HH.MM and press ENTER

if necessary.

Type date and press ENTER. Notice that the computer gives a date and then asks you to enter a date. If the date is correct press ENTER. If the date is not correct, enter the correct date in the format MM-DD-YY and press

ENTER. It is not necessary to enter the day of the week, the computer can determine that from the date that you enter.

Changing the date or time in your computer is easy, but there are a few things to keep in mind. When entering the time, use the 24 hour clock, for example 2PM would be 14:00. You may also find it easier to use a period (.) Instead of a colon

(:), the computer will understand this. It is not necessary to enter seconds or fractions of a second. The date should be entered in the mm-dd-yy format. The month will be first, followed by the day, then the year. All information should be added as two digits. If the date that you wish to enter is July 1, 1990 then enter 07- 01-90.

Syntax: Date allows the user to see the date and to make changes if necessary.

Copy con

The **copy con** command can be used to write small text files to a disk. The file name must be specified and it is also a good idea to specify the drive to where the file is to be saved. After writing the text, **CTRL-Z** must be pressed to save the file. You have just created your first file. In the next exercise we will create several files that we will be able to use in subsequent exercises to practice more DOS commands.

Syntax: copy con [drive]:[filename].[extension]

Once you have typed the text type CTRL-Z to save the file. This command will save a text file to the specified drive with the Specified name.

Type copy con and press ENTER. You will no longer see a DOS prompt.

Now type and press ENTER. Press CTRL-Z to save the file.

List the files on drive a: by typing and pressing ENTER. If the file test.txt did not appear, try again

Filenames

In using the copy con command we have encountered Filenames. Before we create any more files we should have a look at naming files in DOS because DOS has certain restrictions when it comes to naming files and any program that functions with DOS as its operating system must follow the same rules when naming files.

Here are the basics of naming files with DOS

- No two files can have the same name. This applies only to files which are in the same directory of the same disk. (More on directories later)
- The file name is actually divided into two sections. The “filename” and the “extension”.

- The filename can consist of anywhere from 1 to 8 characters in length. You cannot have spaces in your name nor can you have any other of the characters which are used as separators in a DOS command.

The name can be in upper or lower case although DOS will change all the characters to upper case.

- The extension is used quite often to identify a file as belonging to a certain group of documents or

Programs. For example, all your Word-processing files could all be given the extension DOC. This

Would tell you that the file is a Word-processing file. The extension consists of 1 to 3 characters. The

Same rules, in terms of characters which can be used, apply to the extension as applied to the filename.

The extension is optional and, if used, must be separated from the filename by a period. If the extension is not used do not include the period.

Here are some examples of proper filenames:

File1 test-1.his

file2.tst project2.m

1234 proj1020.doc - this is an example of the maximum length that a file name can have.³²

Now that we know how file names work use the command to create the following Text Files on

Drive a:. We will use these files later to practice copying, deleting, moving etc.

Command summary and syntax for lab #1.

Format command: format [drive:]

Switches: command [drive:] /switch

Directory command: dir [drive:]

Changing drives: At the prompt type the new drive letter C:

Label command: label [drive:]

Volume command: vol [drive:]

Version command: ver

Clear screen command: cls

Date command: date

Time command: time

Copy con command: copy con [drive:] filename.[extension]

Subdirectories

Objectives: 1] To become familiar with the different commands dealing with subdirectories.

2] To be able to organize and structure documents and files.

3] To be able to graphically display the structure of a directory.

Up to this point we have learned a few of the basic DOS commands and we have also created several

Files and stored them on the disk in drive a: We are now going to create separate areas on the disk in a: so that we can copy individual files into individual areas. This will help us to organize our files.

Prompt

Before we can start with this process we should have a look at the Prompt command. It facilitates file management in DOS. This command allows you to display any text you want including information such as the name of the current directory, the time, the date and the MS-DOS version number.

Syntax: Prompt [prompt string]

Examples: 1) PROMPT \$Q = (equal sign)

2) PROMPT \$\$ \$ (dollar sign)

3) PROMPT \$T Current time

4) PROMPT \$D Current date

5) PROMPT \$P Current drive and path

6) PROMPT \$V MS-DOS version number

7) PROMPT \$N Current drive

8) PROMPT CALL PETER (Call Peter)

9) PROMPT \$P\$G Current drive, directory and Path

For each prompt example you get a different result or prompt. This can be useful if you want to leave a message to the person using the computer or if you want to leave the time or the

date as the prompt. Also, depending on the prompt that you choose, information such as your current drive, directory and path will be indicated. This is a useful prompt when we are doing file management and it is for this reason that we will leave our prompt as is with the last example.

Make Directory

This command is used to create a directory. **MD** is followed by a space then the name of the new directory. (The rules for naming a directory are the same as those which apply to file names). This directory will be created as a subdirectory of the current directory unless a full pathname is specified.

Change Directory

This command is used to move from one directory to another or to tell you which directory you are currently working in. The **Cd** command by itself shows the current directory of the current drive. To change to another directory, type **cd** leave a space and then type the name of the directory you would like to change to.

Syntax: **cd [drive:][path]**

Cd...

Below are a few examples of the way that the **cd** command can be used to navigate in directories.

Examples:

- 1) **CD** Returns you to the root directory
- 2) **CD...** Moves you one level up in the directory
- 3) **CD\ dos** Moves you into the DOS directory
- 4) **CD\windows\system** Moves you to the windows system directory

Remove Directory

This command is used to delete a directory. To use the **Rd** command successfully you must

- a) Remove all files from the directory which is being removed
- b) Not be in the directory which is being removed when you type the remove command.

Syntax **rd [drive:][path]name**

Copy

The copy command is used to copy files from one location on a disk to another or even from one disk to another. An example of the copy command is shown:

Syntax: **copy [drive:][path]filename [drive:][path]**

Example copy a: document.doc a:\business

The above command would make a copy of the file called **document.doc** found in the root of drive a: and put this copy in the subdirectory called **business**

Wildcards

Copying files using the above method can become time consuming if we have to copy each file one at a time. Using DOS wildcards allows us to copy files in groups. The wildcards are the (*) and the (?). The (*) represents one or more characters and the (?) Represents 1 character.

Syntax: The same as for the regular copy command

Examples: **copy a:*.doc a:\personal** Copies all of the files with the extension doc

Copy a: test1.* a:\personal Copies all of the files with the name test1 and any extension to the subdirectory personal.

Copy a: tes*.* a:\personal Copies all of the files with a name that starts With tes and containing any extension. This file is copied to the Subdirectory called personal.

Copy a: t? St*.* a:\personal Copies all of the files starting with the name T (any letter)st and containing any extension. This file will be copied to the Directory called personal.

Copy a: *.* a:\personal Copies all of the files in the root of a: to the subdirectory personal. As you can imagine, this combination of wildcards is used extensively with the copy command as well as several other commands.

These are only a few of the possibilities which exist using the wildcards. Through experimentation you will become more familiar with their use. The wildcards can be used not only with the Copy command but with the following commands as well: Del;

Dir; Ren; Xcopy; (We have not seen all of these commands as yet but will do so in subsequent lessons).

Delete

The Del command is used to erase files from a disk. This command should be used with care as you may accidentally erase files that you wanted to keep. The command must specify the file name and it is also a good idea to specify the drive.

Syntax: Del [drive:] filename

Examples:

Del a: quest.doc Deletes the file called quest.doc from the root of a:

Del a:\accounts\file1.doc Deletes the file called file1.doc from the

Del a:*.doc Deletes all of the files in the root of a: with the extension doc.

Del a:*. * Deletes all of the files in the root of a:

Be very careful with this combination of wildcards and the delete Command. It will delete all the files in the chosen location. When you use these command dos asks “are you sure”. You must respond y (yes) or n (no) and then press enters.

Command summary and syntax for lab #2

Prompt: prompt

Make Directory md [drive:][path]name

Change Directory cd [path]

Remove Directory rd [path]

Copy Command copy [drive:][path]filename [drive:][path]

Wildcards

Delete del [drive:][path]filename

Other Important Dos Commands

- ✌ Tree command.
- ✌ The chkdsk command.
- ✌ Type and edit commands.
- ✌ Rename command.
- ✌ Dos key command.
- ✌ Extra switches
- ✌ More filter with the pipe symbol.
- ✌ More switches with basic commands.

Tree

The TREE Command is used to graphically display the structure of a directory.

Syntax: tree [drive:] [path] -> the name of the directory where the tree should begin.

Examples:

- 1) TREE (Gives the structure of the default directory.)

- 2) TREE b: (Gives the structure of the disk in the B: drive from another directory.)
 - 3) TREE c:\accounts Gives the directory tree for the subdirectory call accounts on the disk in drive c:
 - 4) TREE /F (Gives the names of the files in each directory along with the directory names.)
- The Pipe character (|) (found on the backslash key on the keyboard):It is used in conjunction with such filter programs as the MORE command (which would pause the output after filling up one screen).

Example: TREE | MORE

Check Disk

This command is used to check the status of your disk and to fix disk errors.

Syntax: CHKDSK [(drive :)(path) filename]

Examples:

- 1) CHKDSK -> Used to display the status of the disk in the current drive.
- 2) CHKDSK B: -> specifies the drive that contains the disk that you want to check.
- 3) CHKDSK /F -> Fixes errors on the disk. Do not use this command from Windows.
- 4) CHKDSK /V -> Displays the name of each file in every directory as the disk is checked

Type

The TYPE command allows you to view the contents of an ASCII file. An ASCII file is one that contains only the characters you see on the keyboard. This command can be useful when we want to see the contents of a file that we created. The files that we created using Copy.com could be files that we verify using Type. This file is used often to view the contents of the Autoexec.bat file used to boot your computer.

Syntax: type [drive:][path]filename

Edit

This command allows you to create text files as well as edit files that are already written.

Note that the “Edit” command is only available in DOS 5.00 and later versions. This command is like the copy con command although it is much more powerful. Using this command you can edit ASCII files.

Syntax: edit [drive:][path]filename

Example: Type: EDIT reminder.txt (Enter) this creates the file

Type: This is an example of (Enter) this creates the text

Type: a text file created using (Enter)

Type: MS-Dos 6's EDIT command. (Enter)

Press: ALT, F, S to save the file

Rename

You may want to change the name of a file if the information it contains changes, or if you simply decide to give it a better or different name. To rename a file, use the REN command.

Syntax: ren [drive:][path]old filename new filename

Example: REN summary.doc report.txt

Example2: REN c:\wperfect\item3.txt item3a.txt

If you want to rename several files at one time the wildcards work with the rename command

Example 1. REN letter.* report.*

Example 2. REN *.doc *.let

Doskey

Doskey (Unfortunately, only available in DOS 5.00 or greater)

The **doskey** command is a handy command to use when working from the DOS prompt. Before using doskey, it must be entered into the memory of the computer.

To do this type doskey at the DOS prompt. Once Doskey has been entered into the memory of the computer it remembers all of the commands that have been entered and these commands can be recalled to be reused or changed. This helps to avoid retyping commands which we use often.

Doskey can be used to find a history of the commands previously used while at the DOS prompt, or to recall commands one at a time. A list of the various doskey commands is shown below:

Key Function

Up arrow recall previous command

Down arrow recall next command in list

PgUp Recall top command in list

PgDn Recall last command issued

F7 Lists all commands in the buffer

ALT-F7 Erases all commands in the buffer

There are other commands, but these will help in our course.

The doskey command and the keys associated with it make working at the DOS prompt an easier process. Instead of retyping commands, commands that have already been used are available for use again or for editing.

F3 function key

This key is similar to the Doskey command although it is not as elaborate a command. With this command you simply recall the last command that you typed at the prompt.

This also can save you of retyping the command that you have just used and need again.

More about Switches

Switches are options which allow us to do more with certain commands. With a couple of the commands we have discussed thus far we have looked at switches available with these commands. In the following section we are going to have a look at a few commands which we have already discussed but with which we will now also use switches. To discover which switches are available for a given command, type the command followed by /?

Example DIR /?

Some examples of DOS commands used with switches:

Example 1: DIR /P displays one screen of listing at a time

Example 2: DIR /W use the full width of the screen

Example 3: DIR /O displays a listing of directories first alphabetically - then a listing of files alphabetically.

Example 4: DIR /O/P combining 2 switches in same command

Example 5: DIR /S display a listing of all files in the present directory as well as in all subdirectories.

This can be used to find a specific file. DIR WP.EXE /S

Example 6: COPY a:\project.txt c:\proj1.doc /v to verify the copy action to see that it was copied correctly.

Example 7: DEL letter.doc /p to confirm that you really want to delete this file before actually doing it.

Example 8: Copy a:*.* /y will prompt to confirm if you want to overwrite an existing destination file.

Command Summary and Syntax for Lab #3

Tree tree [drive:] [path]

Check Disk chkdsk [[drive:][path]filename][/f]

Type type [drive:][path]filename

Edit edit [drive:][path][filename]

Rename ren [drive:][path]old filename new filename

Type dir /?

Type dir c: /p

Type dir c: /w

Type dir c: /o

Windows XP Installation

It is important to understand that this guide was specifically designed for a lab environment. There are a lot of operating system vulnerabilities that are intentionally left unpatched in these installation steps. This is intentionally done to give you the best results when completing the labs and tutorials in this book. If you are interested, a great reference for building a Windows XP Professional box that is secure enough for a production environment is *Windows XP Security: Step By Step* by SANS.

To create a properly configured laptop for the Security Essentials Boot Camp, follow the detailed steps in this document—from the initial setup screen to the final login. This guide was designed for use on a system that doesn't already have a Windows platform installed on it. If your machine does not have a blank hard drive, some of the screens you see at the beginning of the installation may be different from what you see in this chapter. If different screens appear, it is important that you always choose the option to *replace*, or *overwrite*. Do not choose to upgrade. The Windows install should also be placed in the default c:\windows directory.

Creating Boot Disks

If your system does not support the capability to boot off of a CD-ROM, you can use the Windows XP boot disk to boot. If you do not have a set of the four disks, you need to use a machine that already has Windows XP Professional installed on it. The following steps show you how to create the four boot disks:

1. Label four blank, formatted, 3.5-inch, 1.44-MB floppy disks as: **Setup Disk One, Setup Disk Two, Setup Disk Three, and Setup Disk Four.**
2. Insert **Setup Disk One** into the floppy disk drive of a Windows or DOS system.
3. Insert the Windows XP CD-ROM into the CD-ROM drive.
4. Click **Start**, and then click **Run**.
5. In the **Open** box, type **D:\bootdisk\makeboot a:** (where **D:** is the drive letter assigned to your CD-ROM drive), and then click **OK**.

6. Follow the screen prompts.

7. After you have completed the screen prompt requests, insert **Setup Disk One** into the floppy disk drive of the lab PC and power the PC on.

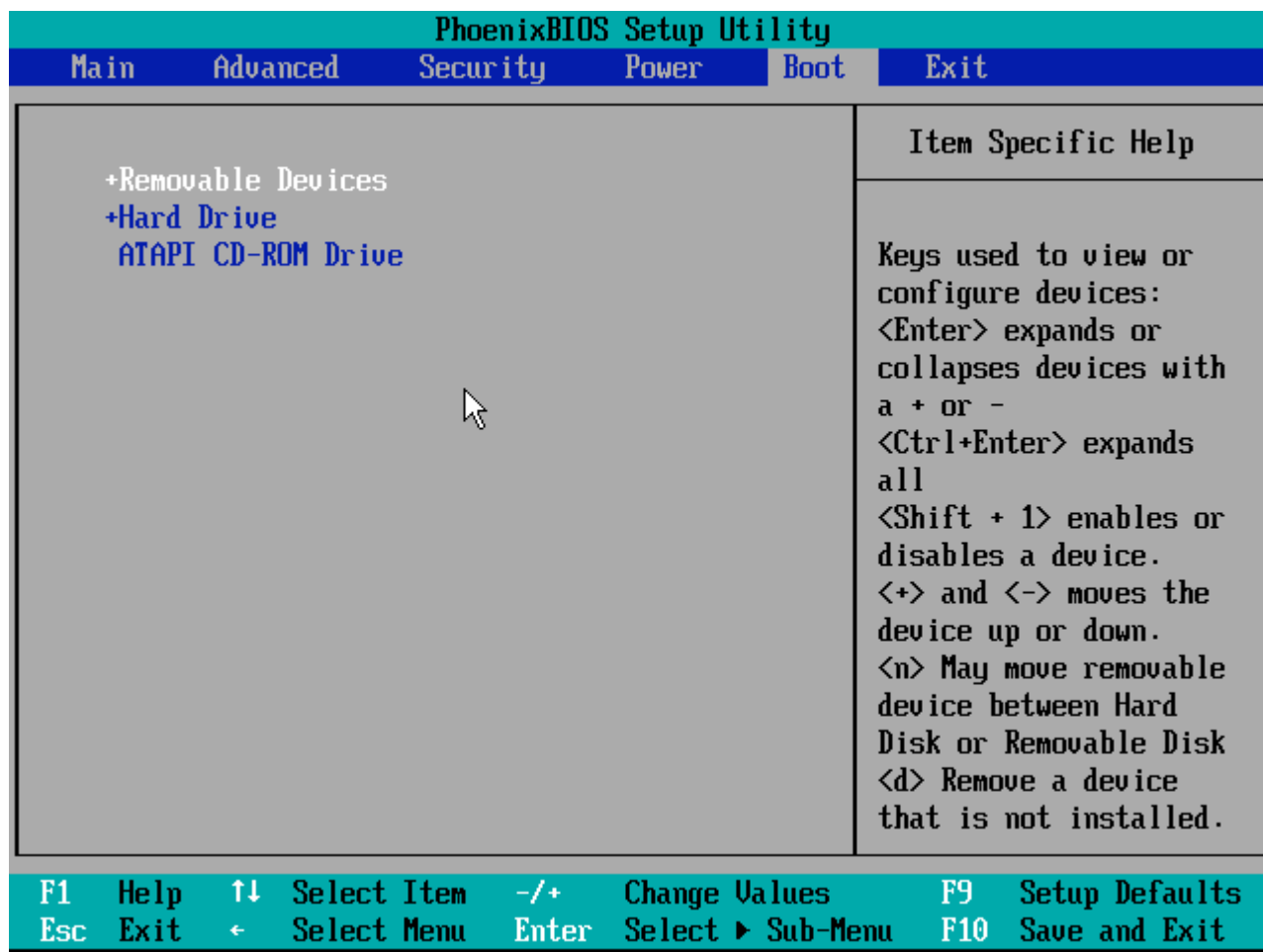
Booting from the CD-ROM

If your system supports booting off of the CD-ROM, you do not need to use the disks previously discussed. Instead, follow these steps:

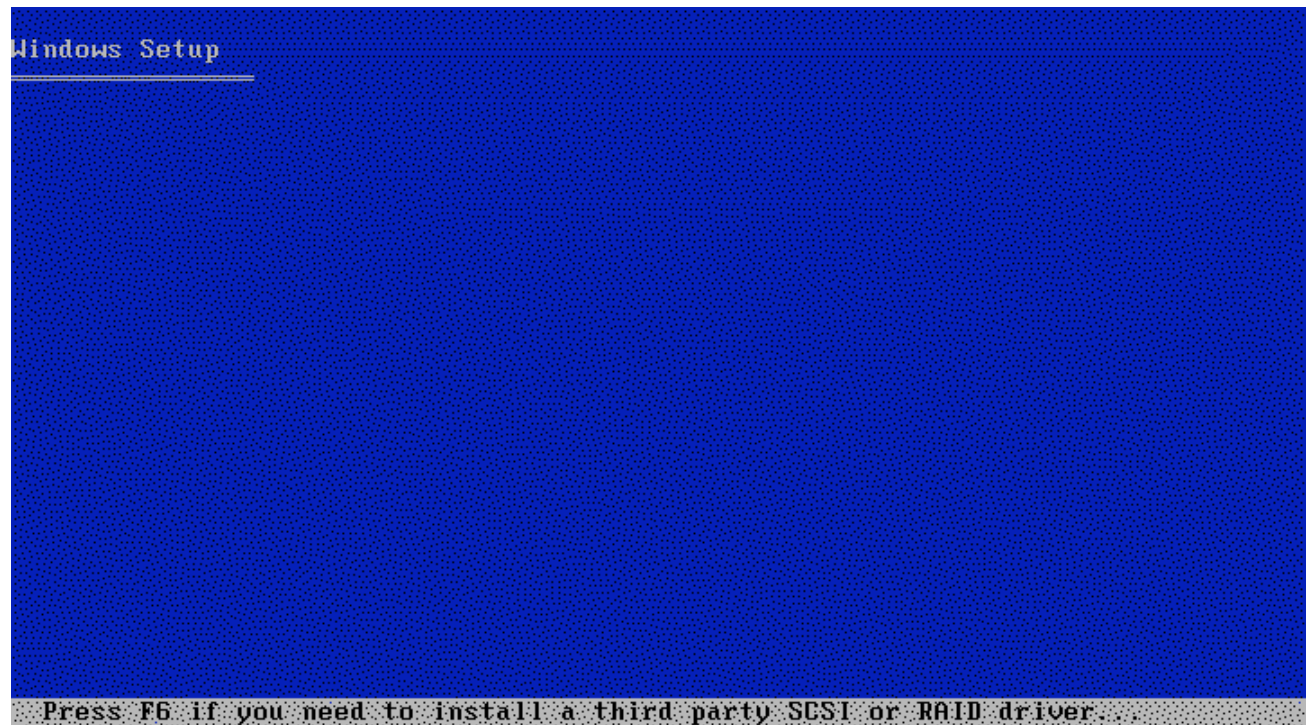
1. Simply start by placing the Windows XP CD-ROM into your CD tray and power on your machine. The first non-blank screen you should see is the one shown in the following illustration.



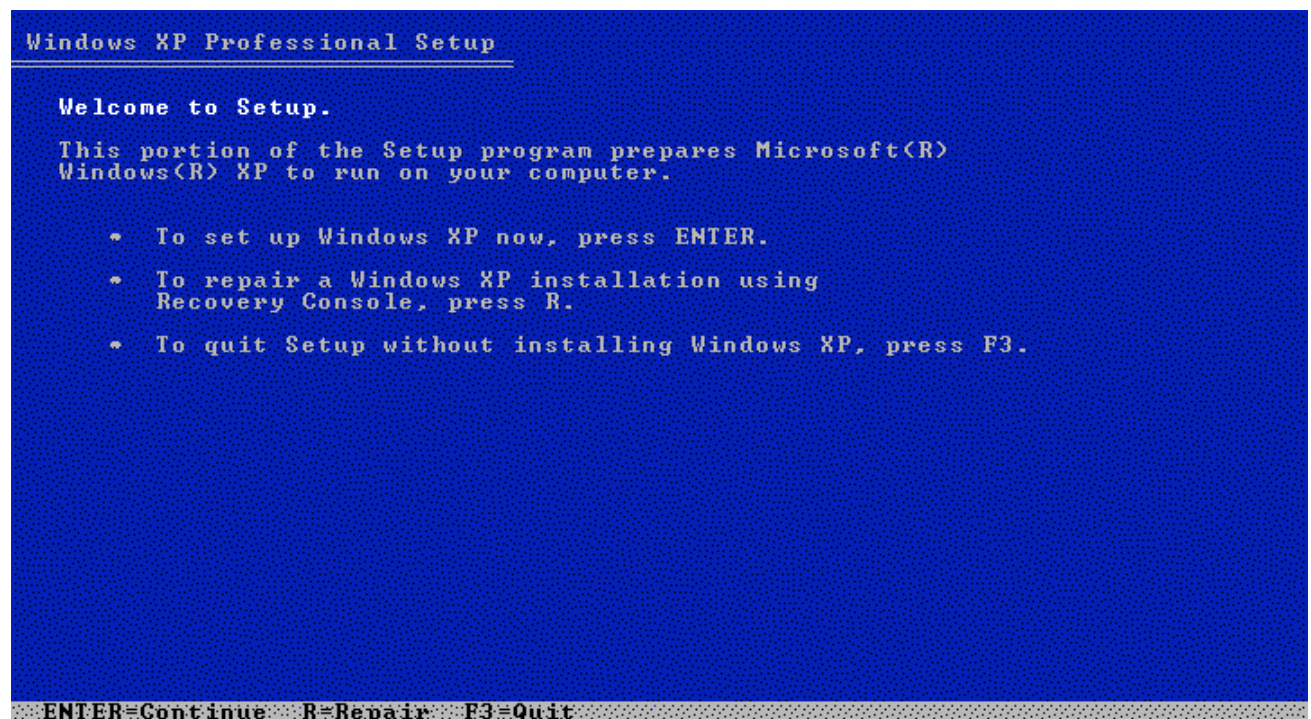
2. If the previous screen does not appear, reboot your machine and open up the BIOS. You need to make the system boot to the CD-ROM first. The following screen is one of several different BIOSes you could have on your system. You need to navigate to a screen that allows you to change the Boot Order. This is where you tell it to boot off of the CD-ROM.



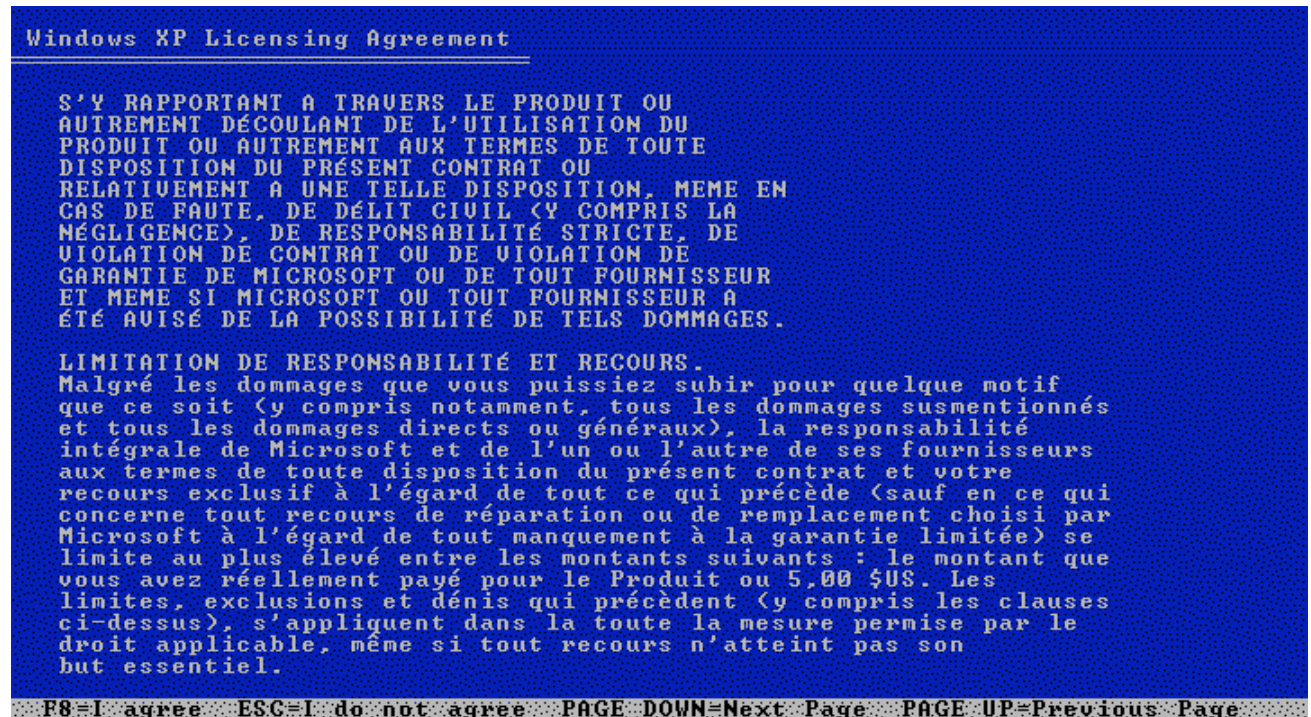
3. Now your system should boot off of the CD-ROM. After a period of time (typically 30-45 seconds), the following screen appears. Because we are doing an initial install, you only need to press **Enter** to continue.



4. Hit **Enter** at the next screen to continue installation.



5. The Microsoft Windows XP Licensing Agreement appears next, as shown in the following screen. It is important that you read and understand this agreement before continuing with the installation. After you have read and agreed to the contents of the license, press **F8** to continue.



Defining Drive Partitions

You now need to define the drive partitions. Defining your drive partitions is used instead of FDISK. When defining your drive partitions, it is extremely important that you leave enough space for your Linux partition! Following are the steps:

1. Press **C** to create a partition for your Windows install.
2. You need a minimum of 2 GB of space for each of your operating systems.

When you are prompted for the size of the partition, enter a number that is equal to 50 percent of your available hard drive space. Then, highlight the partition, which should be labeled **Unpartitioned space** (see the following illustration), and press **C**.

Note: If partitions already exist they should be deleted. However you should realize that this will permanently remove any data that is currently on your system.

```
Windows XP Professional Setup

The following list shows the existing partitions and
unpartitioned space on this computer.

Use the UP and DOWN ARROW keys to select an item in the list.

  * To set up Windows XP on the selected item, press ENTER.
  * To create a partition in the unpartitioned space, press C.
  * To delete the selected partition, press D.

8190 MB Disk 0 at Id 0 on bus 0 on atapi [MBR]
  Unpartitioned space 8189 MB

ENTER=Install C=Create Partition F3=Quit
```

Now create your new partition to be at least 2 GB. In the provided space type **2047** and Press **Enter**

```
Windows XP Professional Setup

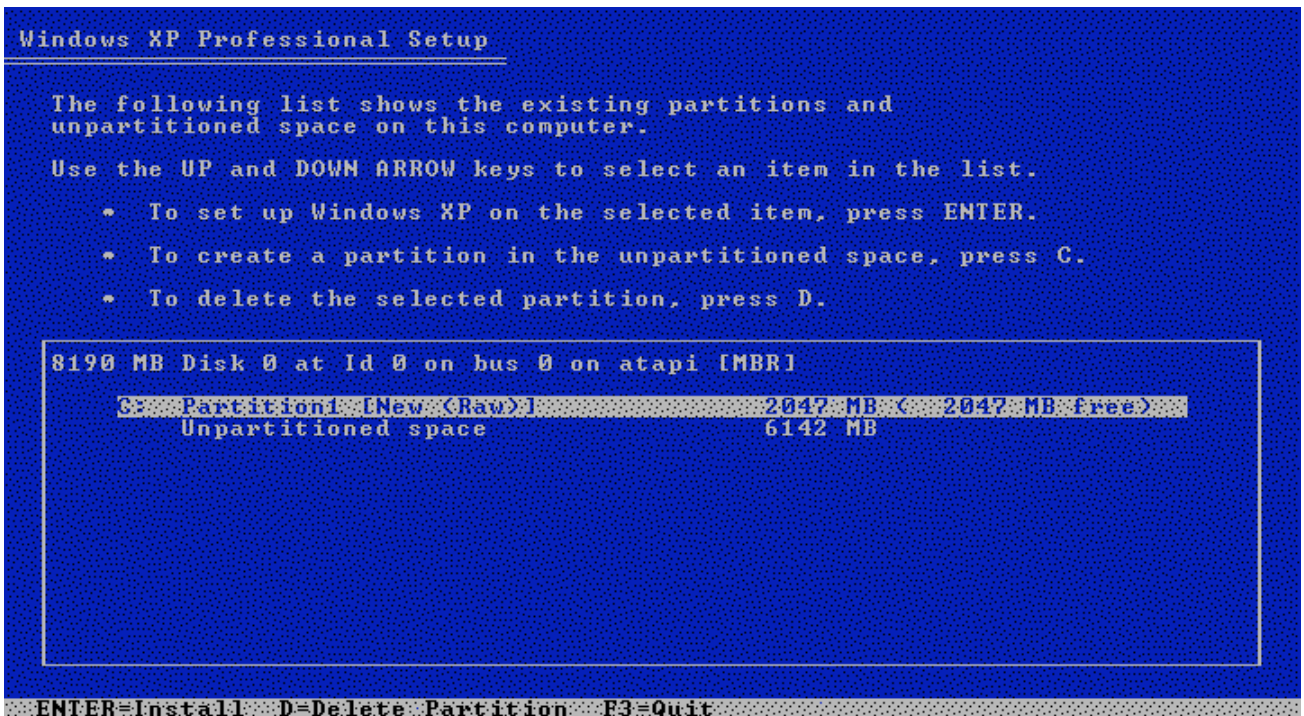
You asked Setup to create a new partition on
8190 MB Disk 0 at Id 0 on bus 0 on atapi [MBR].

  * To create the new partition, enter a size below and
    press ENTER.
  * To go back to the previous screen without creating
    the partition, press ESC.

The minimum size for the new partition is 8 megabytes (MB).
The maximum size for the new partition is 8182 megabytes (MB).
Create partition of size (in MB): 2047

ENTER=Create ESC=Cancel
```

You should now see two partitions. Verify that the new 2047 partition is highlighted and press **Enter**.



Formatting Drive Partitions

The next step is to format your partition. For security reasons, you should format your partitions using NTFS. NTFS is a Windows partition type that allows you to assign permissions at the folder level. This level of granularity is not the same for FAT partitions. NTFS also allows for larger partition sizes compared to the 2 GB limit that comes with FAT16. The steps for formatting your partition follow:

1. Highlight the NTFS <Quick> partition option as shown in the following screen, and press **Enter**.

Windows XP Professional Setup

The partition you selected is not formatted. Setup will now format the partition.

Use the UP and DOWN ARROW keys to select the file system you want, and then press ENTER.

If you want to select a different partition for Windows XP, press ESC.

~~Format the partition using the NTFS file system (Quick)~~

Format the partition using the FAT file system (Quick)

Format the partition using the NTFS file system

Format the partition using the FAT file system

ENTER=Continue ESC=Cancel

2. After you press **Enter**, the system formats the partition, as shown in the following screens. Depending on the size of the partition, this step can take from 5 minutes to an hour. (You may need it because you have a long way to go.)

Windows XP Professional Setup

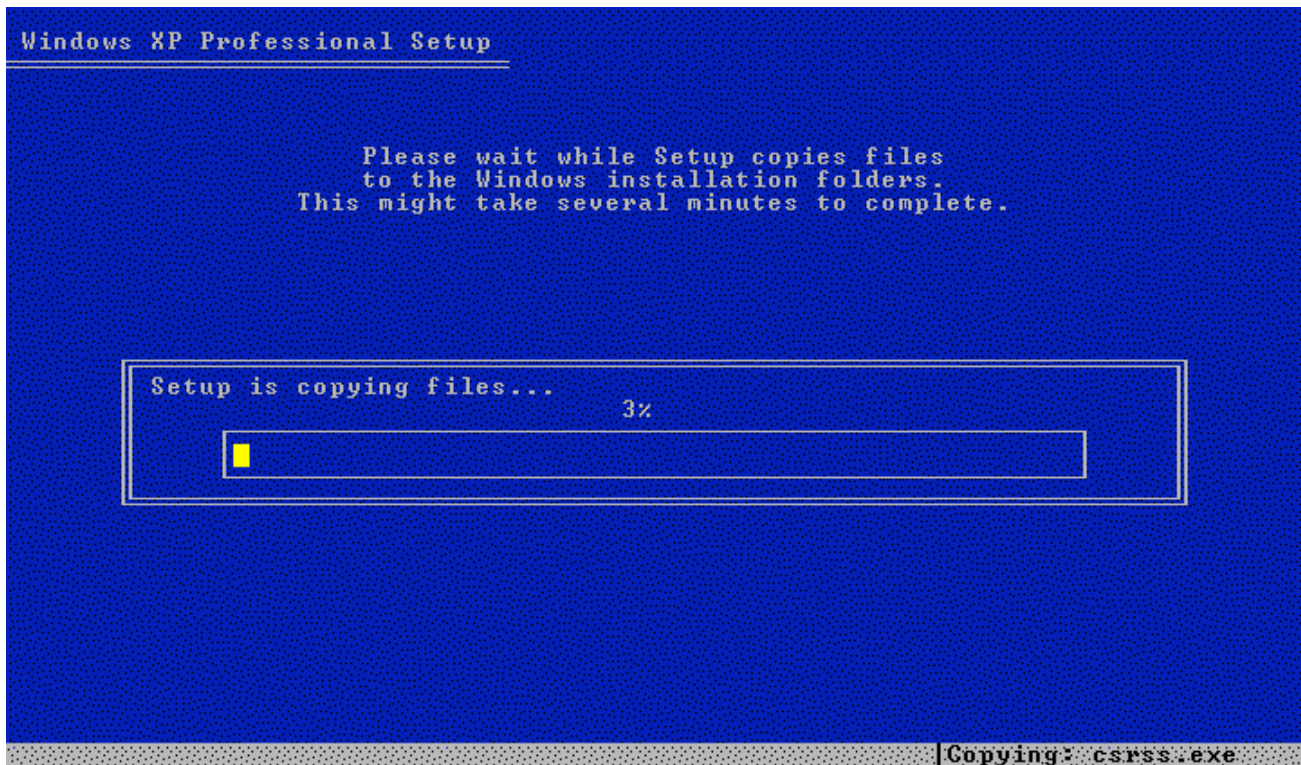
Please wait while Setup formats the partition

C: Partition1 [New <Raw>] 2047 MB < 2047 MB free>
on 8190 MB Disk 0 at Id 0 on bus 0 on atapi [MBR].

Setup is formatting...

0%

Since this will take a while you should just wait while this process continues.

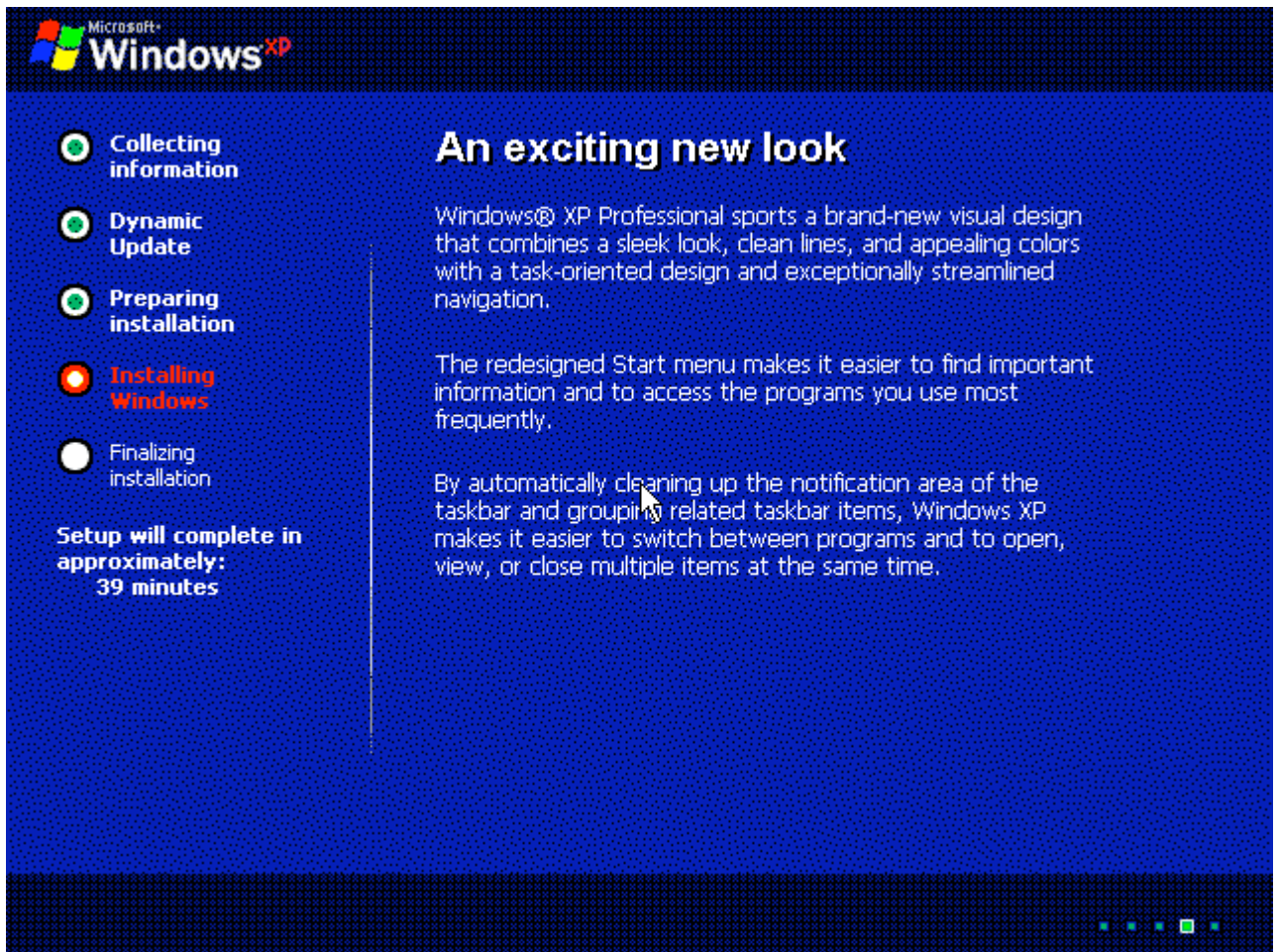


When you return to your machine, you may see one of the following screens. Don't be alarmed. The system has completed the formatting process and has automatically rebooted. After this occurs, you have to answer the remaining install questions.



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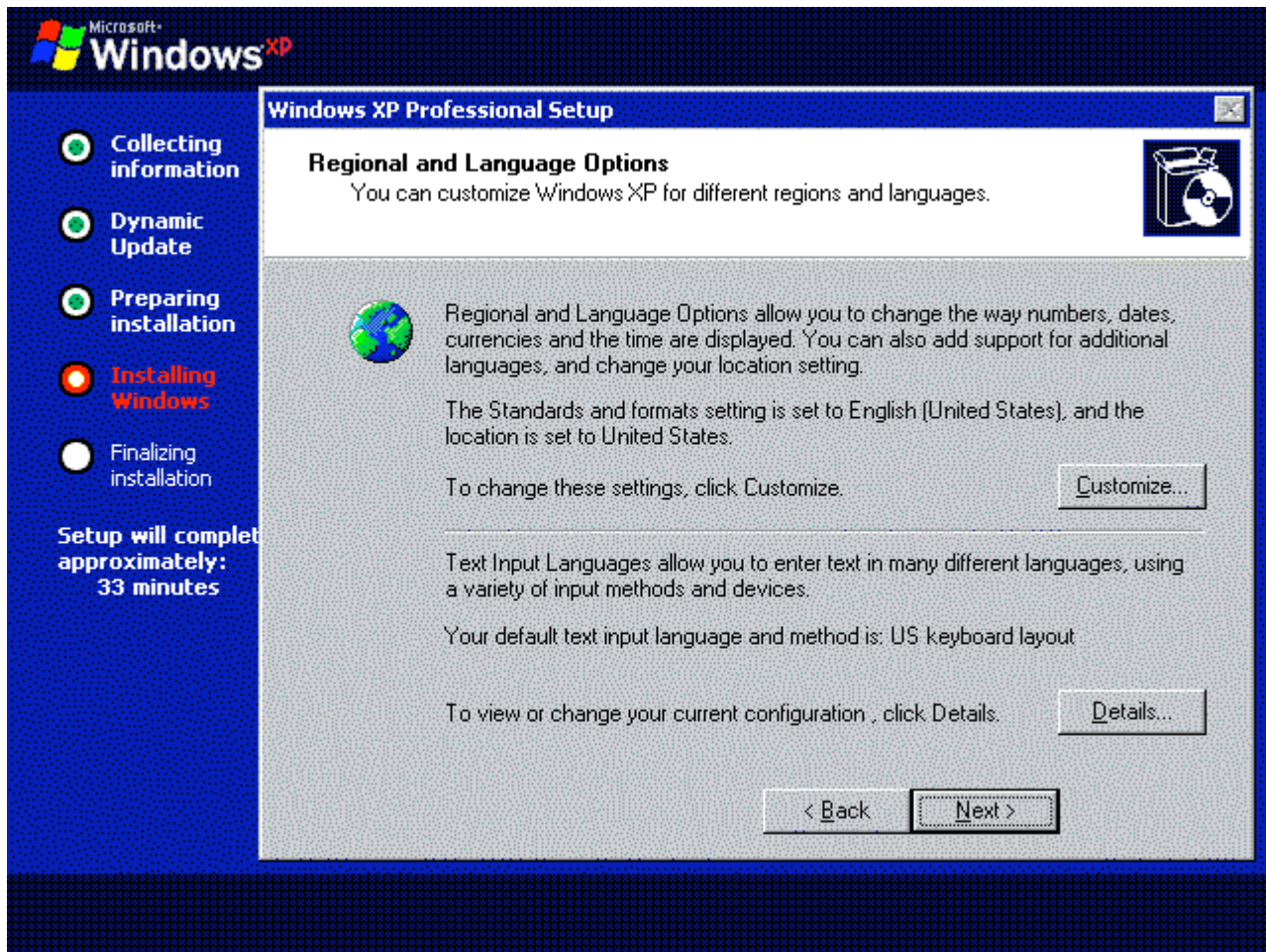
Microsoft



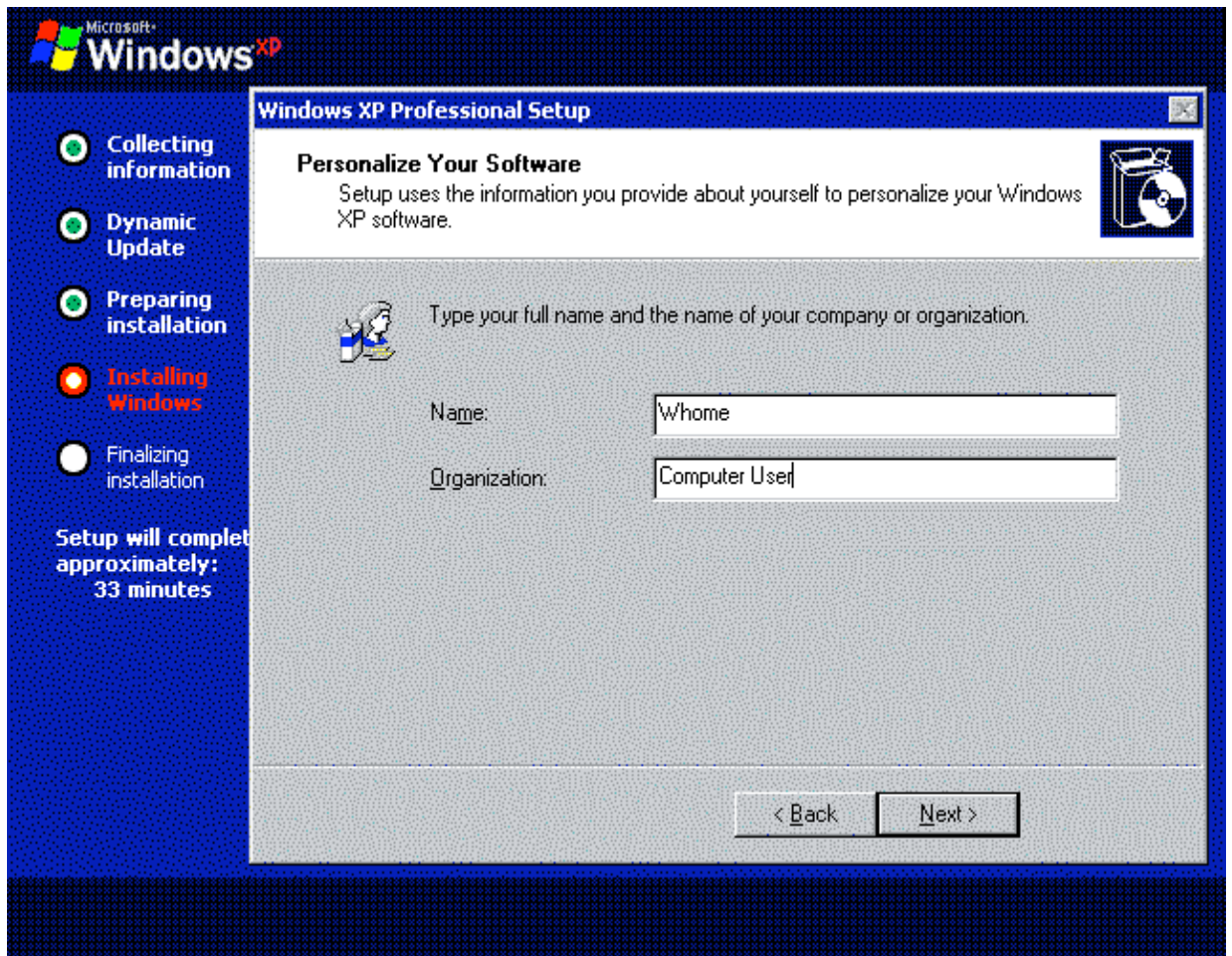
Customizing Your System

Now Windows presents a series of questions, which, when answered, customize your system. The following steps walk you through the process of customizing your system:

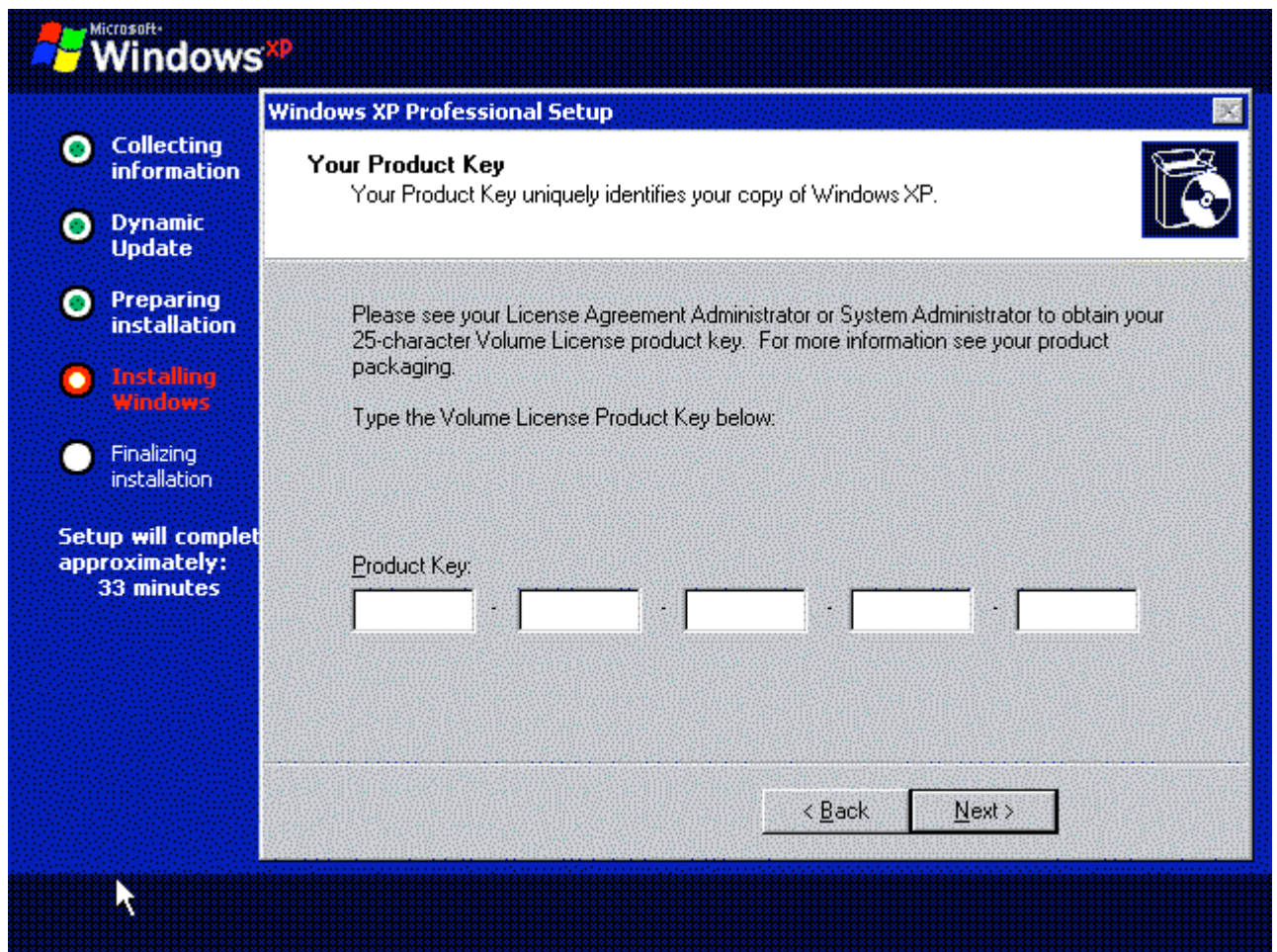
1. Typically, you only need to make changes during the next step (see the following screen) if you are located outside of the United States or if you use a non-standard keyboard. If you are in the United States and you are using a standard QWERTY keyboard, press the **Next** button. If you are located outside of the United States, you should change your locale settings.



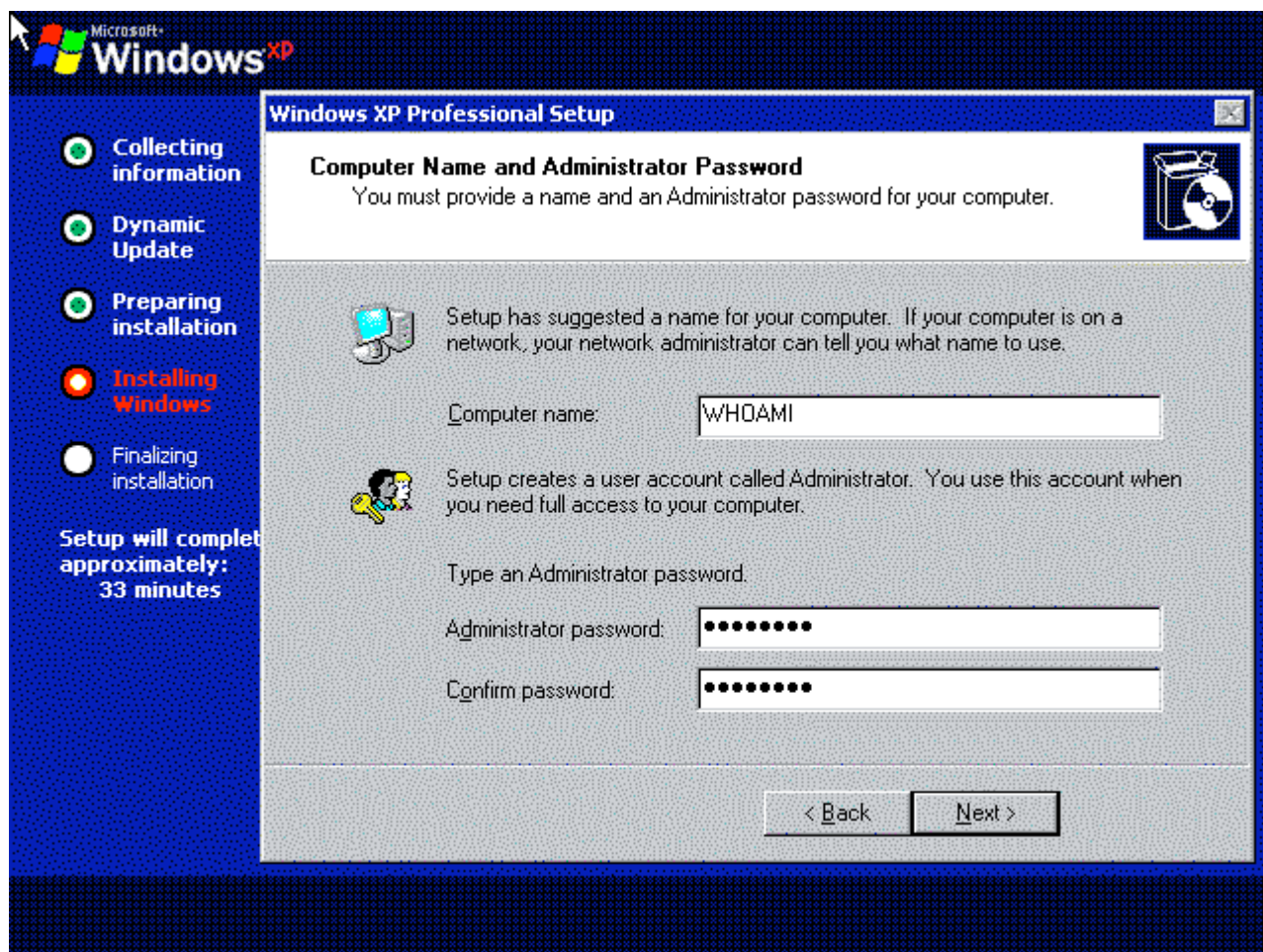
2. Enter your name and the organization you work for in the **Name** and **Organization** fields. For the purposes of this course, have some fun making up fictional names. Click the **Next** button when you are done.



3. In the next screen, enter the Product Key number that came with your software (find it on your CD). If you make a mistake when you enter then key, you receive an **Invalid Key** message and the system gives you another opportunity to enter it. Once you enter in the valid key, press the **Enter** key.

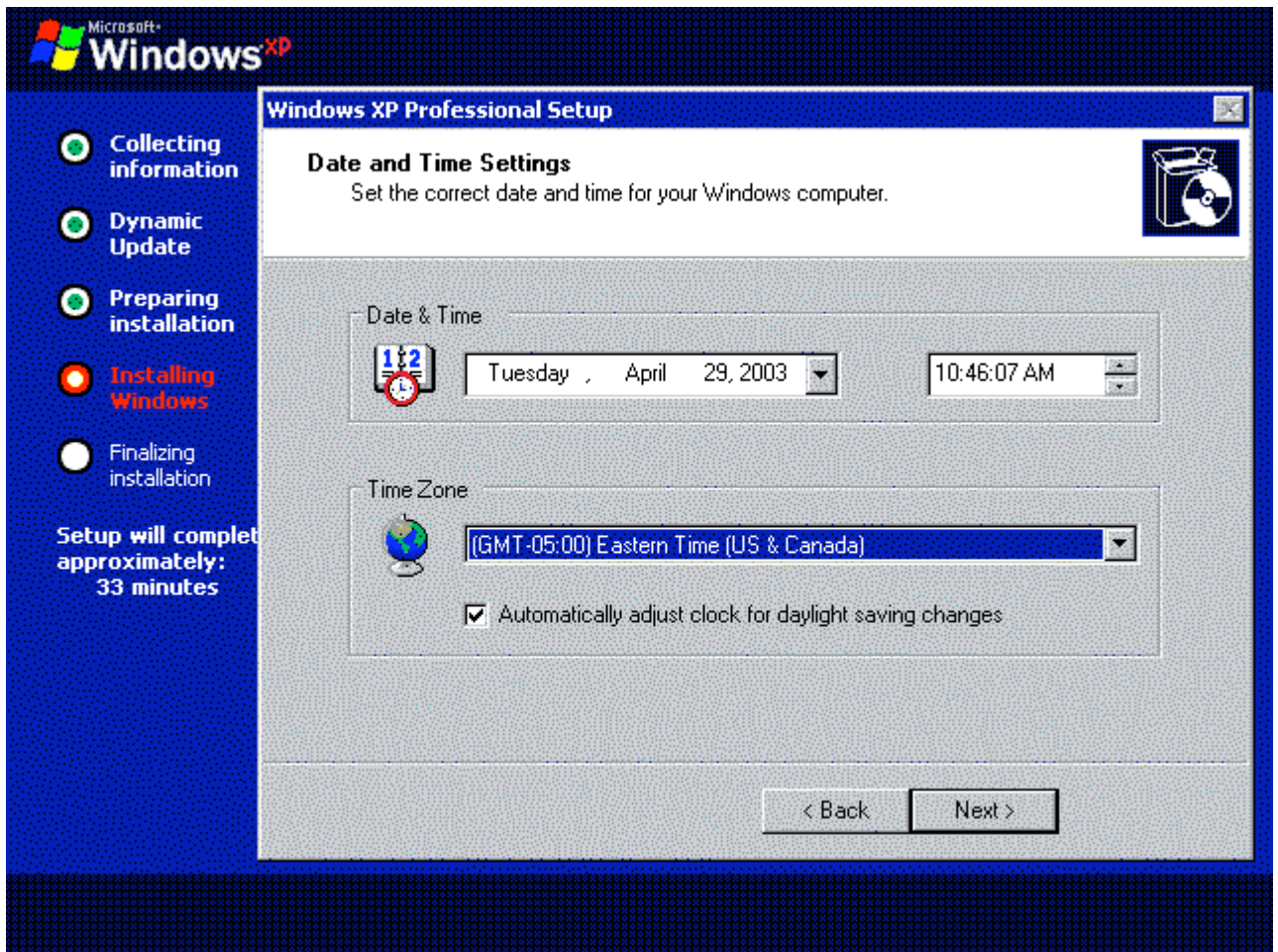


4. Now enter a name in the **Computer name** field to name your computer. If you are part of a corporation's domain, you need to follow your corporation's guidelines for naming systems. For our purposes, name your machine whatever you desire. Then, type in a password in the **Administrator password** field. You also need to confirm the password, as shown in the following screen. Then, click the **Next** button. **Warning:** A common mistake many administrators make at this stage is to leave the **Administrator password** field blank. It is highly advisable that you enter a password that matches your company's password policy for local passwords. You don't want to forget to change the password after you have completed the installation. Also, make sure you remember this password. You will need it to login.

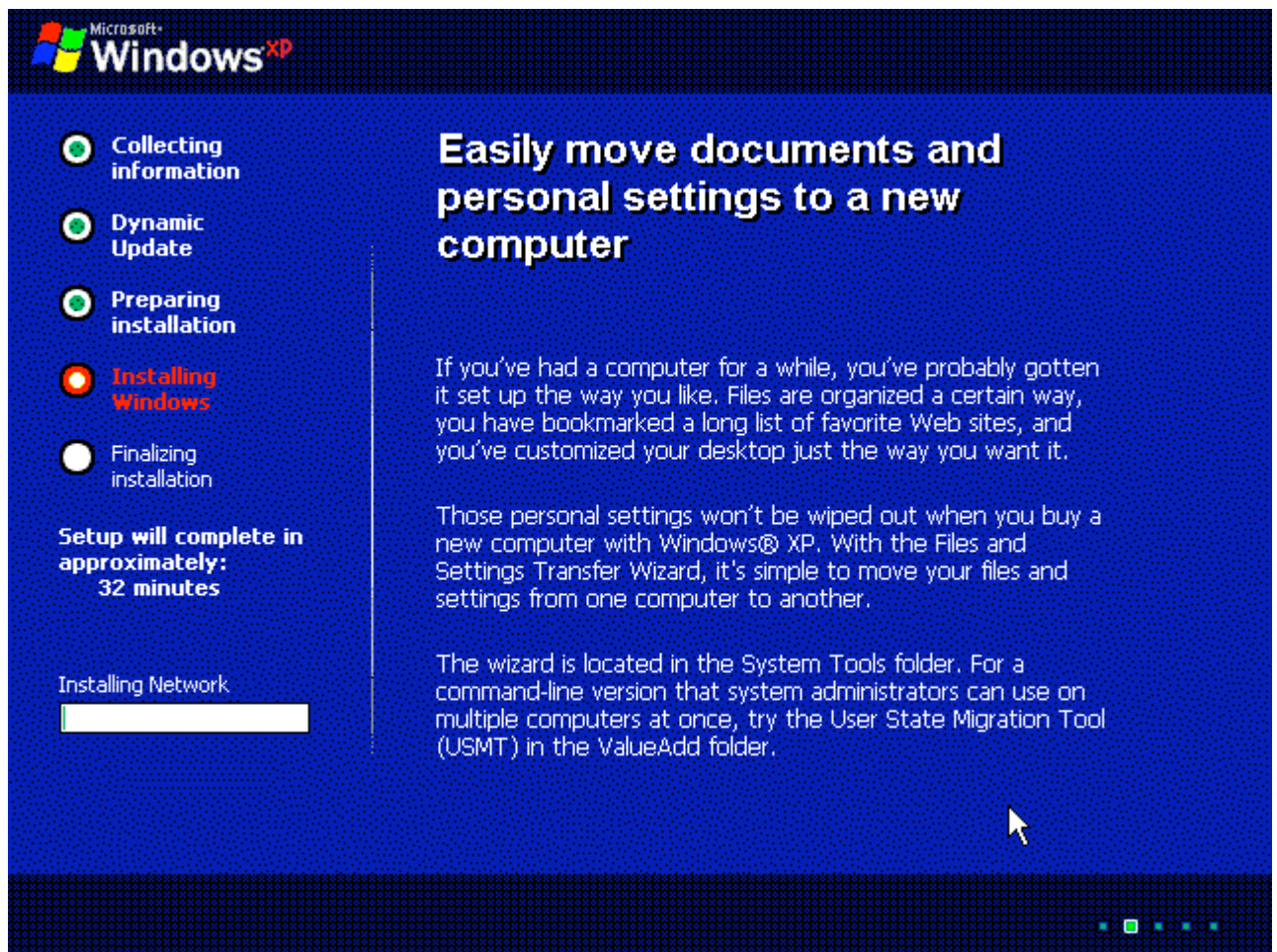


Note: Depending on your configuration, you might receive the Modem Dialing Information Screen. Just cancel out of this or click Next to get to the next screen.

5. In the screen that appears, enter the current time, and then fill in the **Date** field and **Time Zone** field. Click **Next**.



6. After you make the previous configurations, the system installs your networking components, as shown in the following screen.

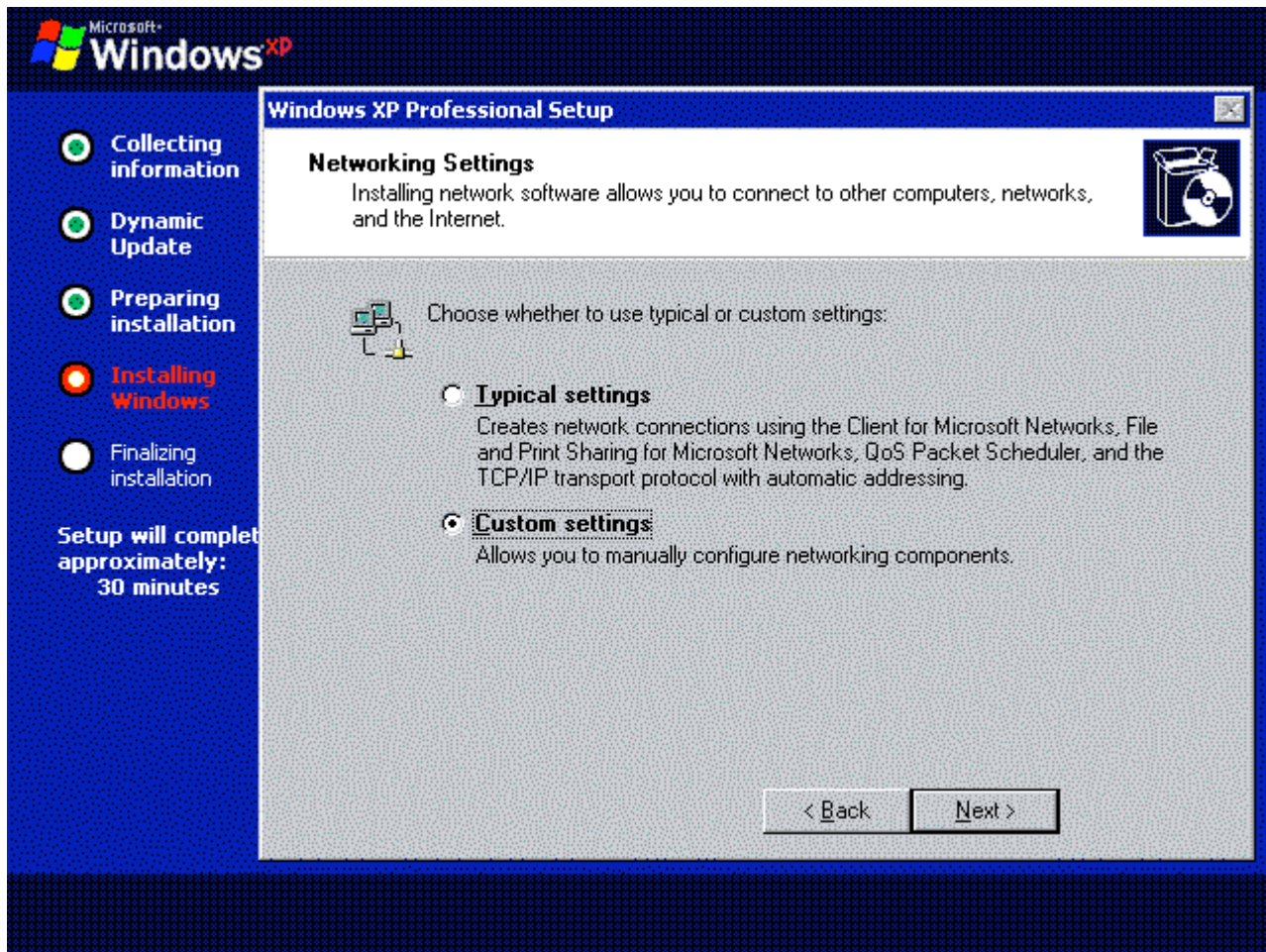


Customizing Network Settings

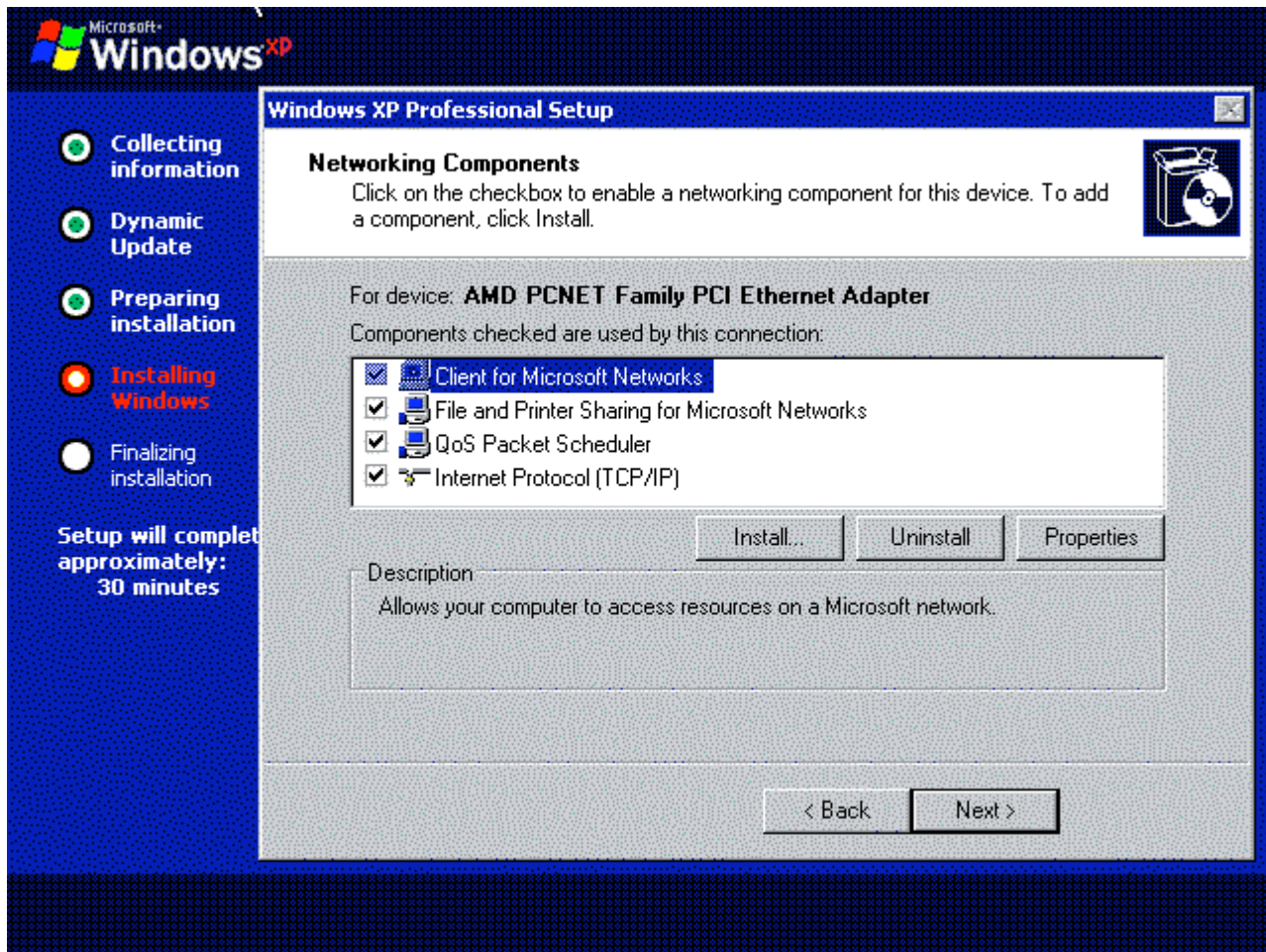
Now you need to set up your system so that it can be networked with other systems.

Following are the steps:

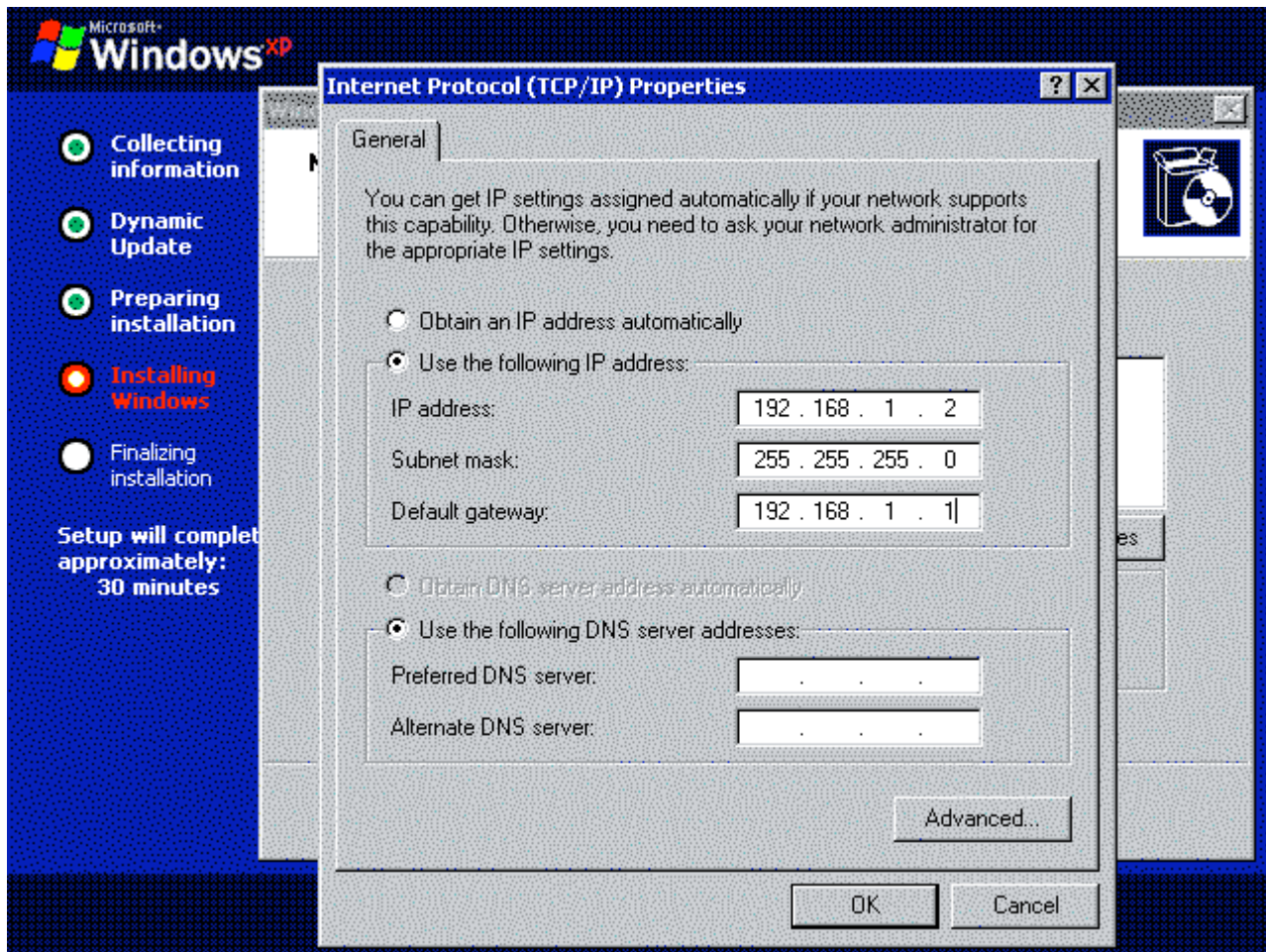
1. First, you must choose the type of settings you are going to use. Note that it is rarely a good idea to use **typical settings** when configuring an application or operating system. It is always a good idea to choose **Custom settings**, as shown in the following screen. When you use this option, you only install options you need. You won't end up installing something you aren't aware of because you chose an option that automatically does this. After you select the **Custom settings** option, click **Next**.



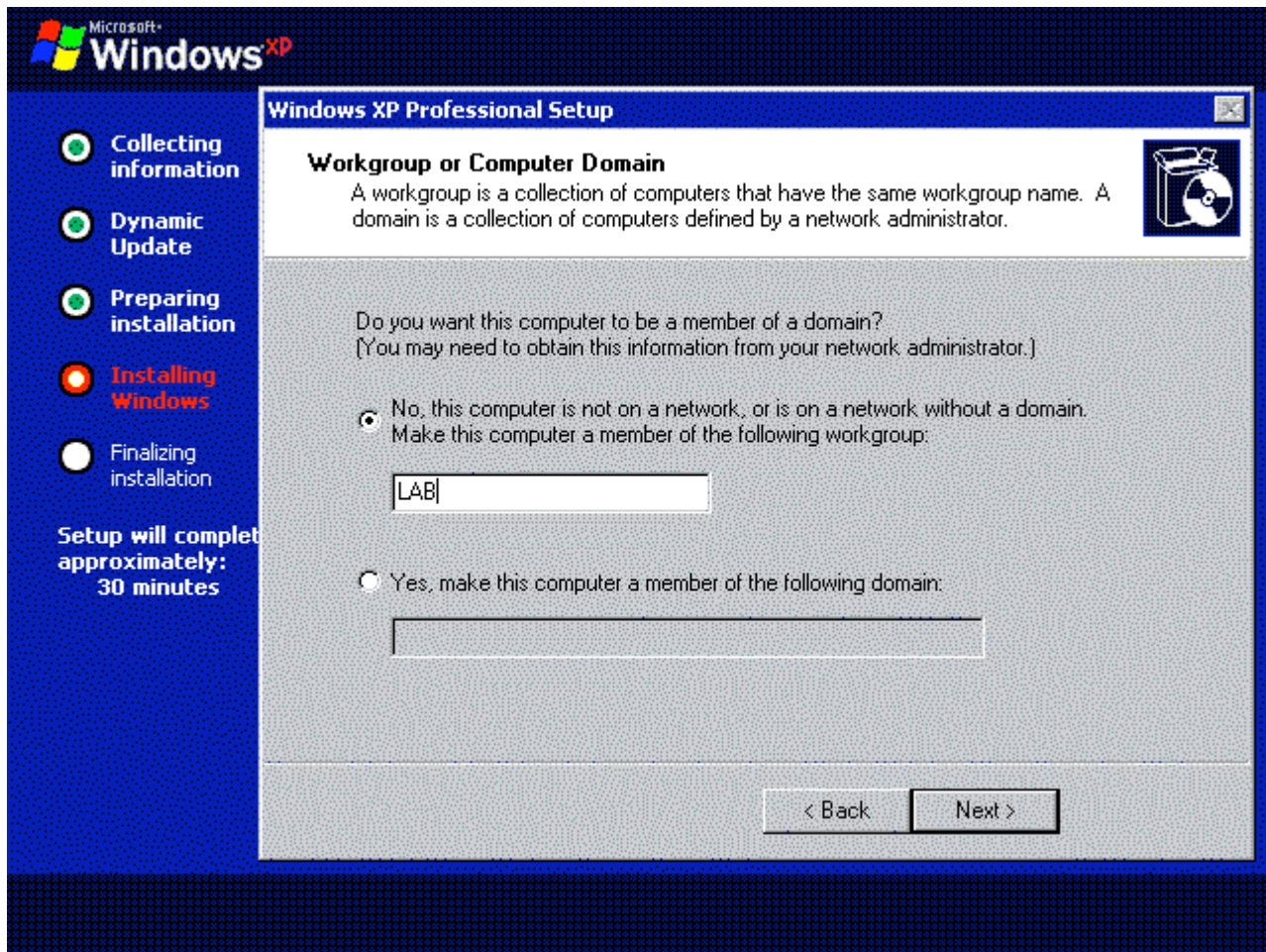
2. Windows no longer tries to install IPX/SPX, so there is nothing in the custom settings that you need to remove. This is a great time to setup your local IP address if you are not using a DHCP server in your environment. The assumption here is that you are not going to plug this test machine into a production environment, so it's safe to add your own IP address. Highlight **Internet Protocol (TCP/IP)** and click the **Properties** tab.



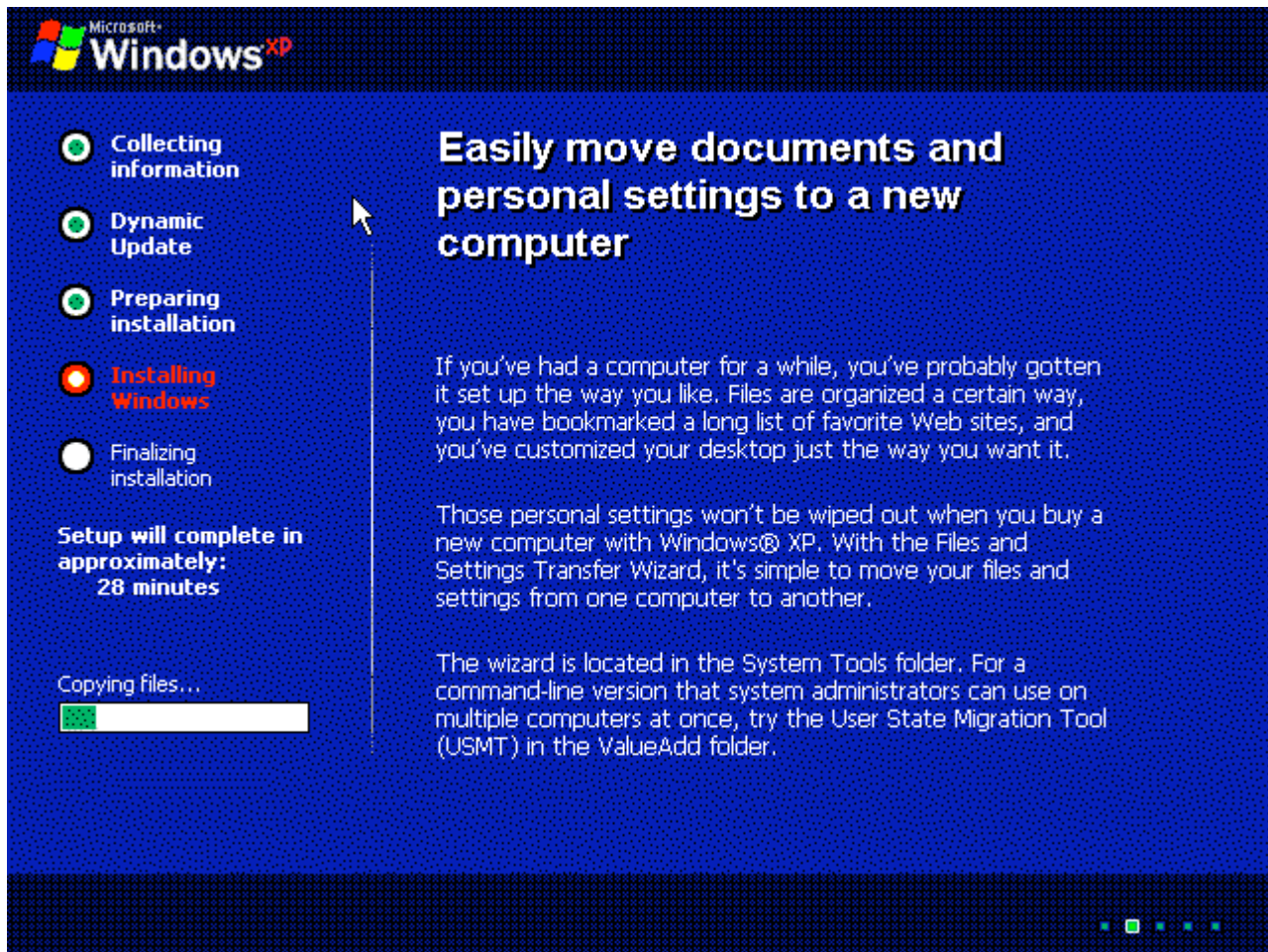
3. The following screen appears which allows you to enter your own IP address. For the purposes of this exercise, use a non-routable IP address. Select 'Use the following IP addresses. Enter **192.168.1.2** in the correct fields. Then, enter a standard 24-bit subnet mask of **255.255.255.0**. To make the entire section complete, enter a default gateway setting of **192.168.1.1**. Enter the appropriate DNS server IP addresses for your environment into the fields shown in the following paragraph. You can leave your DNS sever fields blank for this system. Click OK. Click Next.



4. As previously stated, you are not joining a network or a domain, so just enter a name of your choice and leave the first **No** option enabled (see the illustration that follows). After you have the information entered, click Next.

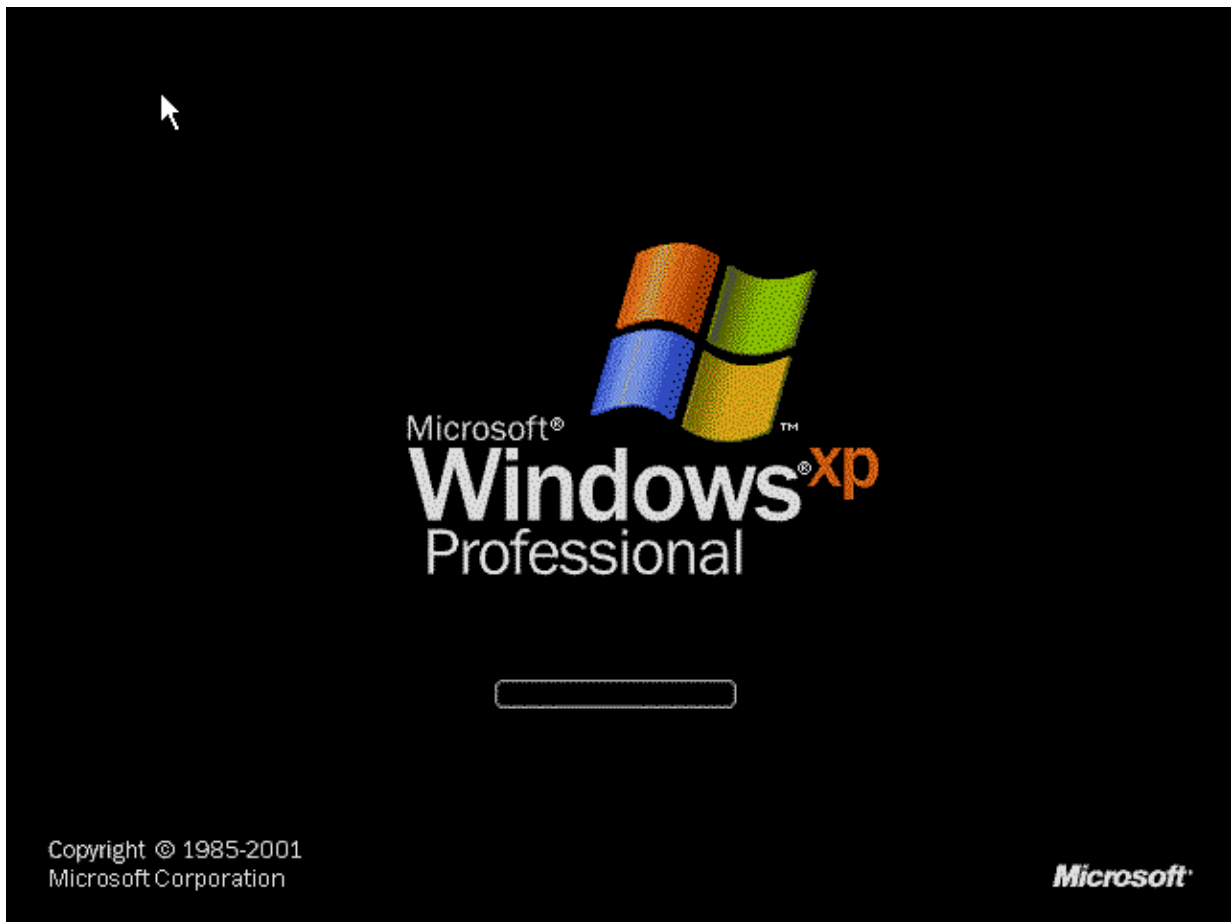


5. Windows completes the networking portion of the installation and moves on to its final tasks. This step takes a long time, so take the opportunity to grab another caffeine-laced beverage.



6. If you get the following screen, shout for joy. Congratulations, you have successfully installed Windows XP. Click **Finish**, and then remove the Windows CD-ROM before the system reboots so that you don't

Accidentally start the install process again. If you accidentally leave the CD-ROM in, and the install process starts again, simply remove the CDROM and hard-boot the machine (restart it).

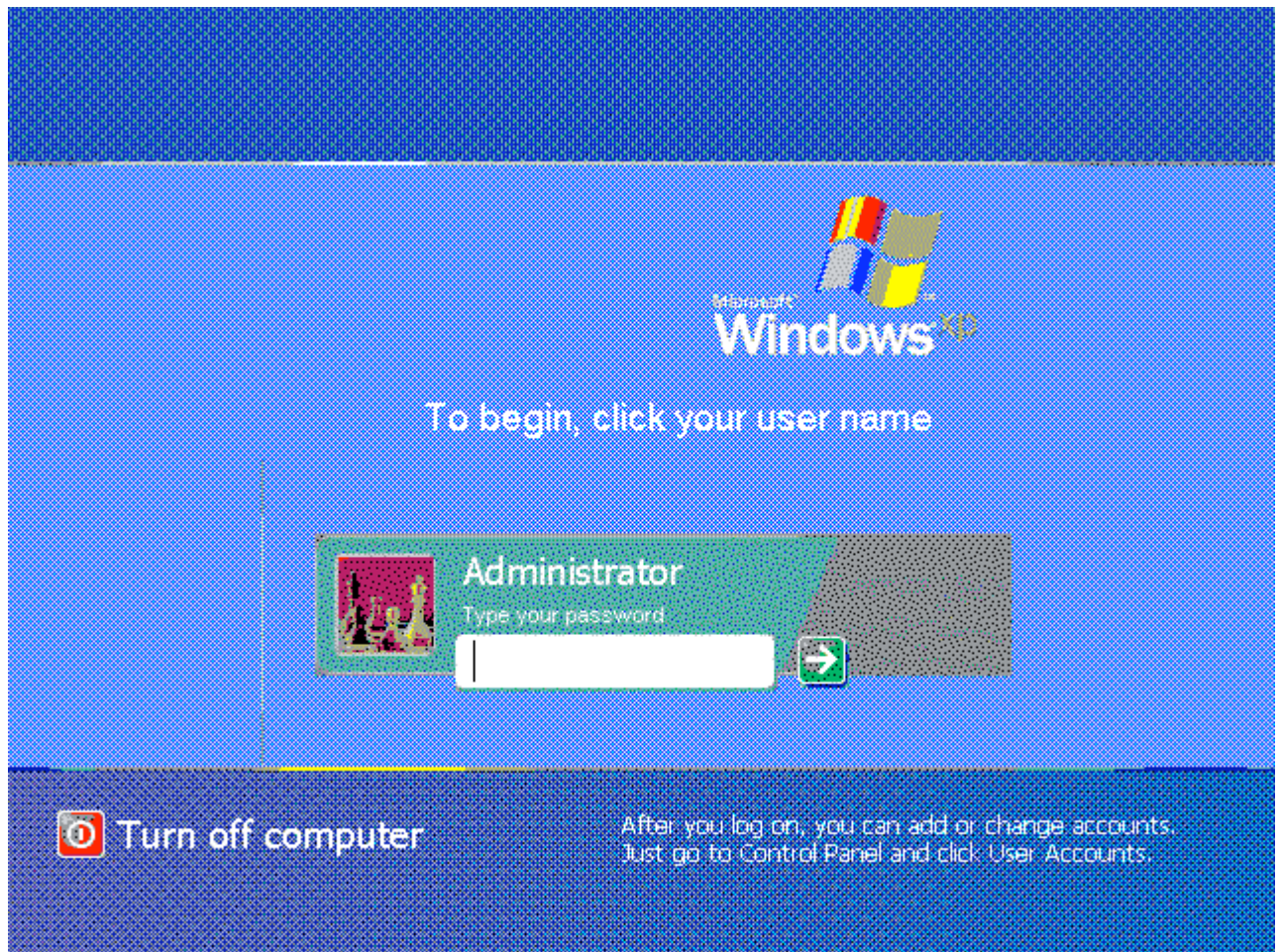


9. After the next screen comes up, click the **OK** button.



10. As shown in the next screen, you now need to log in using the Administrator account and the password you entered earlier during the install. After you have entered the appropriate credentials, click **OK**.

Note: Depending on your version of the software you might get several screens about connecting to the Internet and registering Microsoft before you get the login screen.



Note: At this stage, the base installation of Windows XP is installed. The instructions that Follow show you how to upgrade to Service Pack 2, which is recommended for this class.

Installing window 7 operating system

Windows 7 Installation Step by Step

In order to install your copy of the Windows 7 operating system please makes sure to have the following items:

-Windows 7 Installation disk

-Drivers CD: * Use after installation process if needed.

***NOTE:** Please make sure to back up all of your important data before proceeding with this installation.

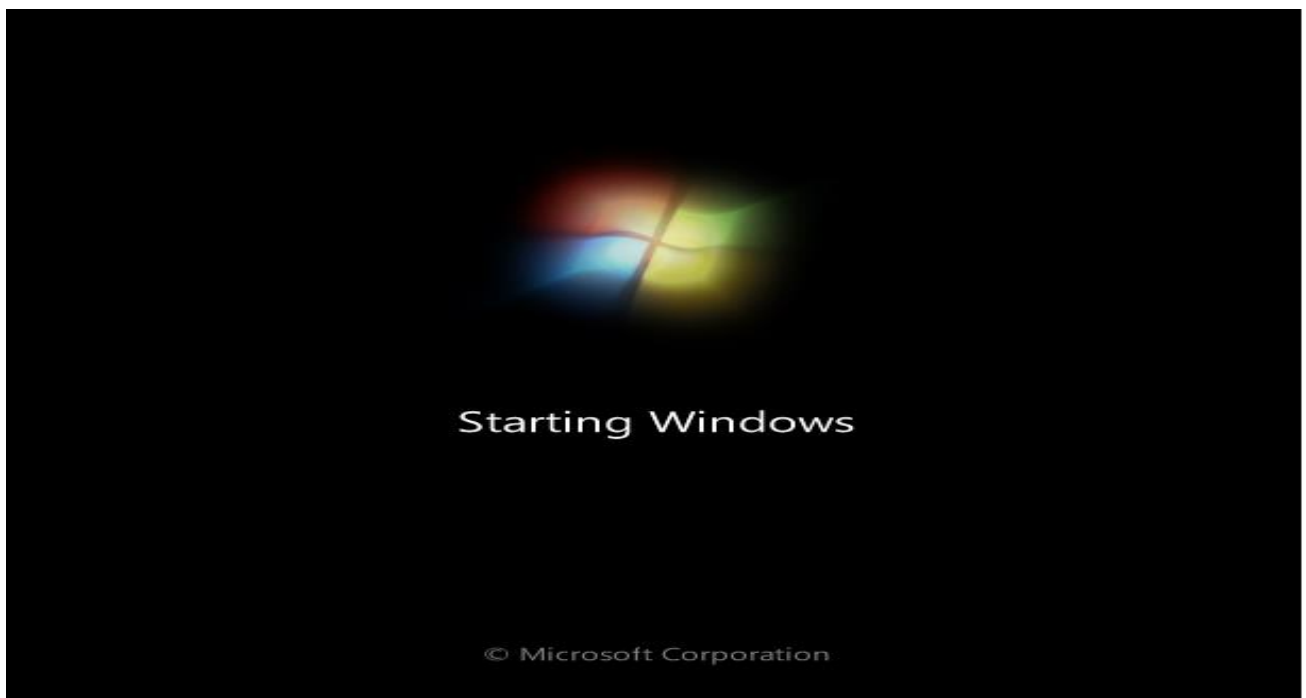
Step 1. Insert the Windows 7 operating system disk into your DVD drive, and then restart your computer

Step 2. You will see a prompt that says '**Press any key to continue**' after the ZT logo disappears. When

you see this press any key immediately.

***NOTE:** The following screen shots may differ from what you may see during your installation. (i.e. Win7

version, HDD disk space etc..)



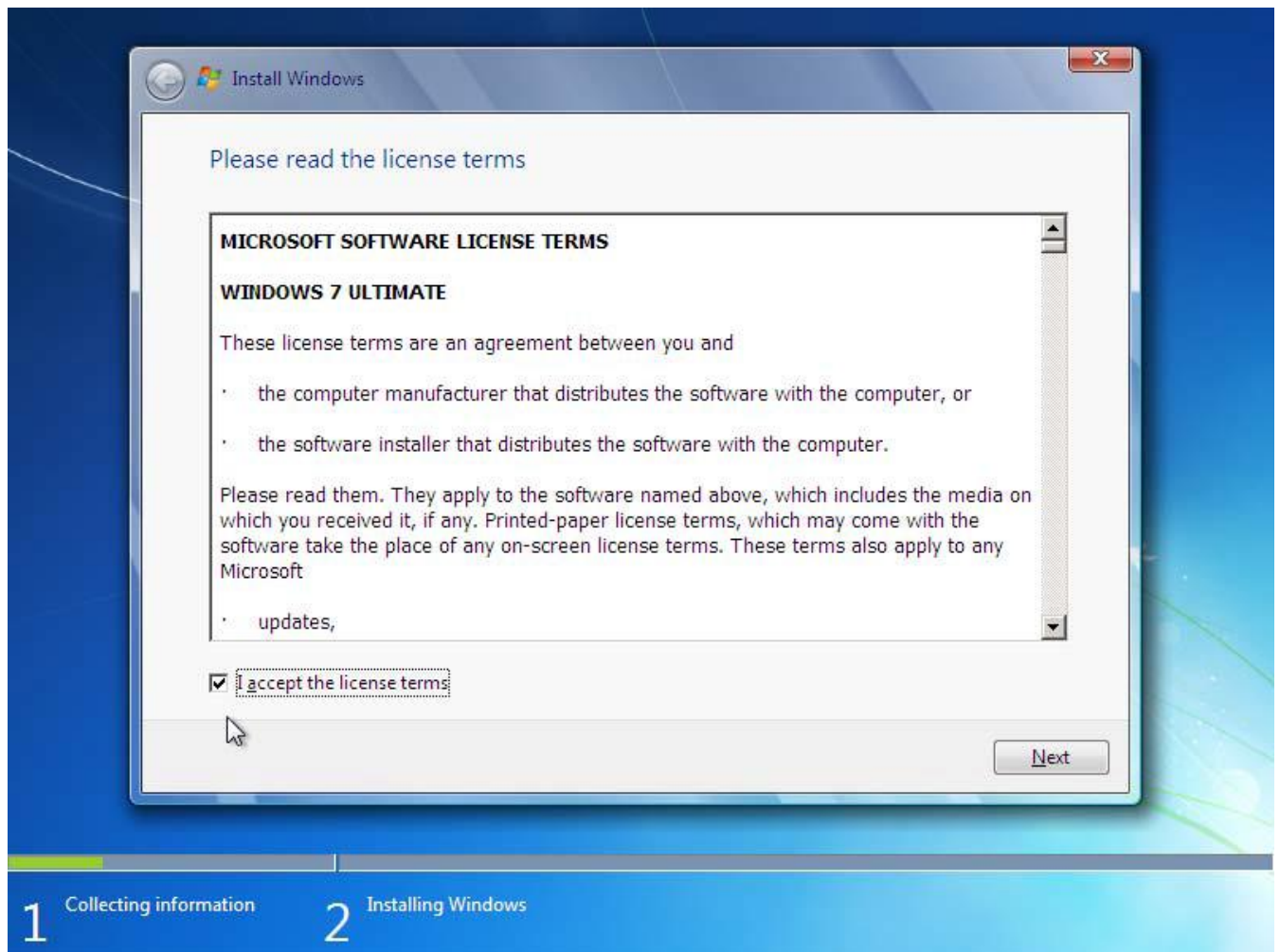
Step 3 '**Starting Windows**' with the Windows7 logo will appear.



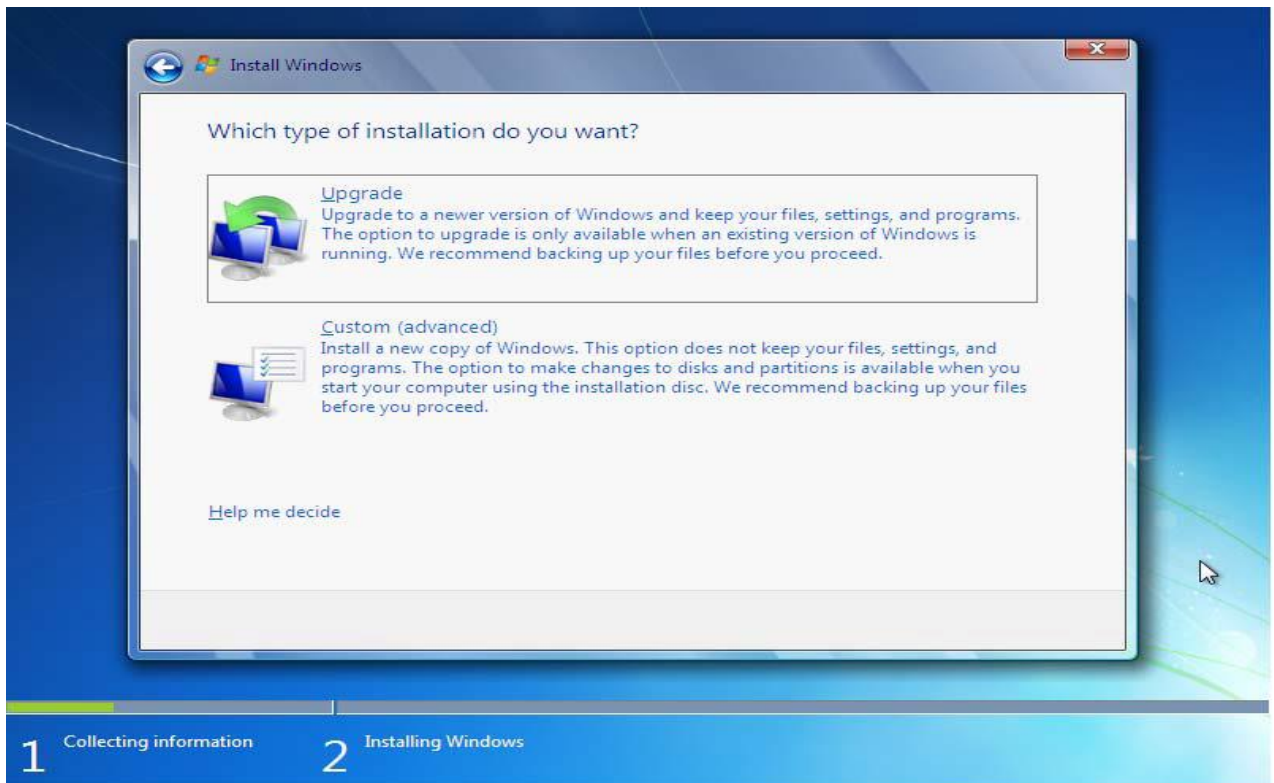
Step 4. Language options, by default English will be set along with “time and currency format” and “keyboard or input method”



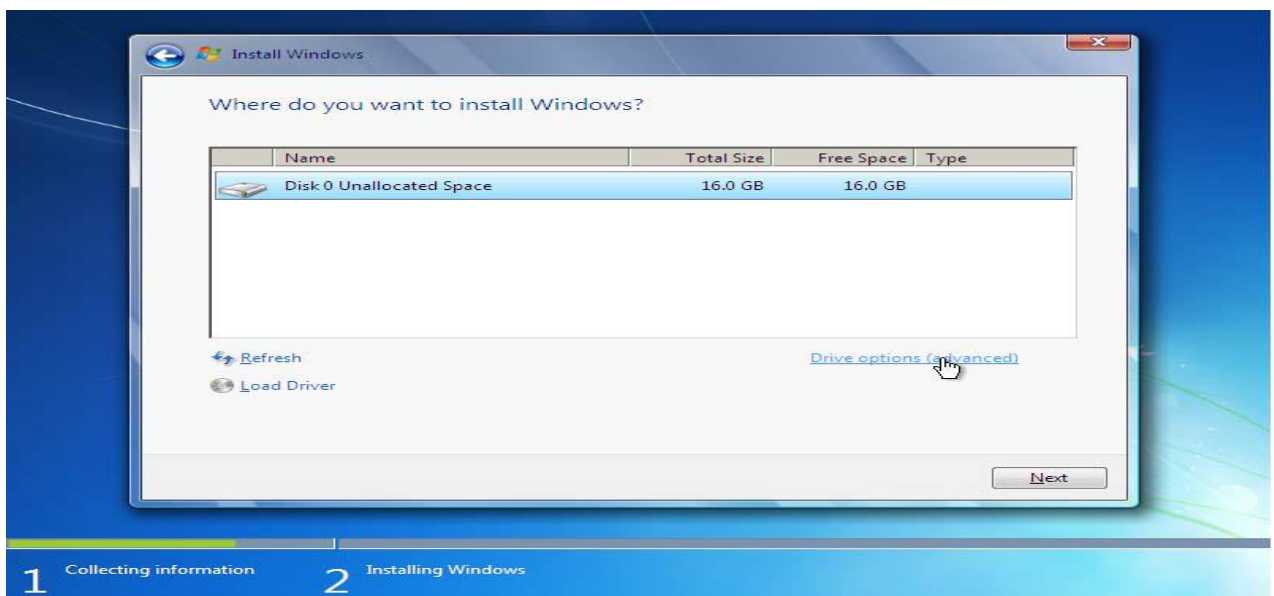
Step 5. Click “Install Now”



Step 6. End User License Agreement (E.U.L.A.), check the box to accept, and click “Next”



Step 7. “Which type of installation?” window will appear. Upgrade will be grayed out; the only option you should be able to choose is Custom (advanced).

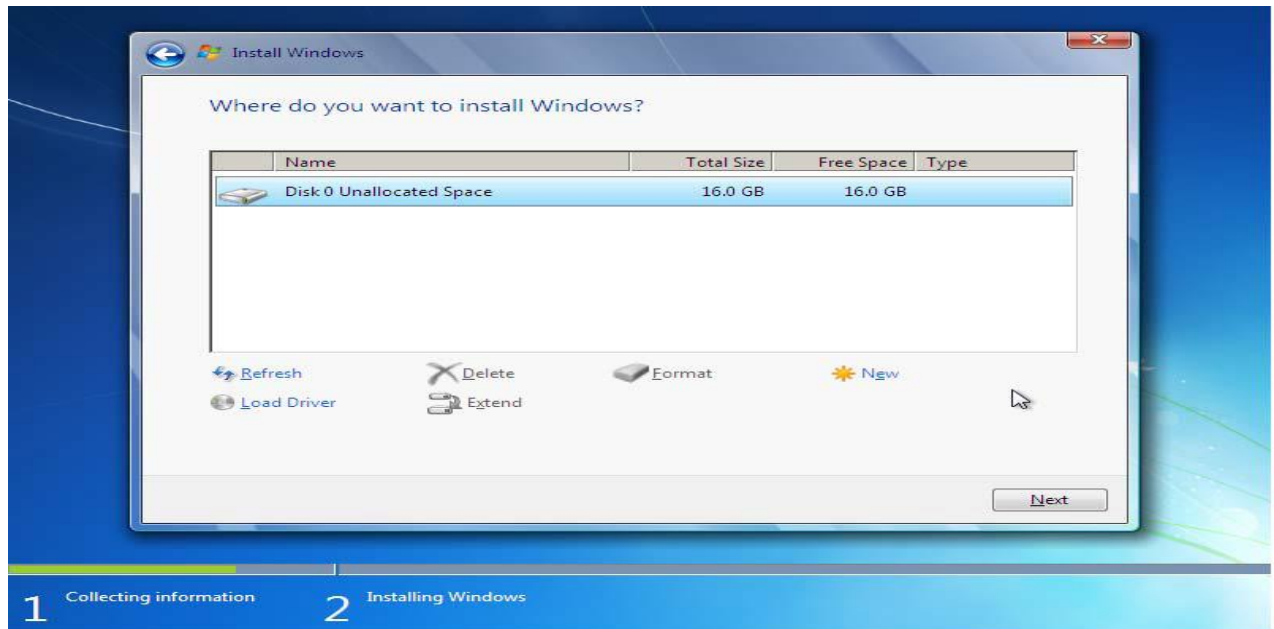


Step 8. “Where do you want to install windows?” Make sure the partition is highlighted.
Step 9. Delete the partition by clicking on Drive options (advanced) on the bottom right corner of the field.

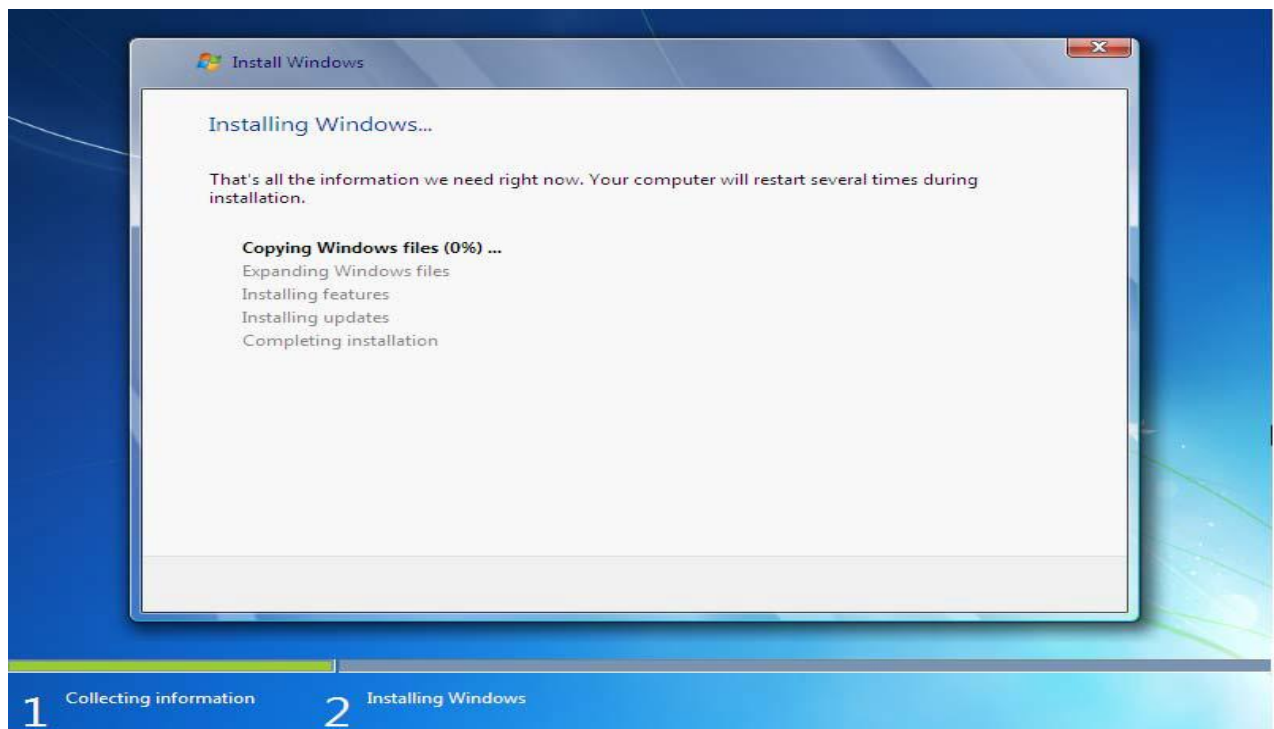
Make sure the partition is highlighted and click on Delete. If drive advanced options is grayed out, then the Partition will not have to be deleted.

***NOTE:** Deleting the partitions will erase all data on the system

Step 10. Disk 0 Unallocated Space should be the only listing at this point. If it is press next, If not please proceed to delete any additional partitions that may be listed.



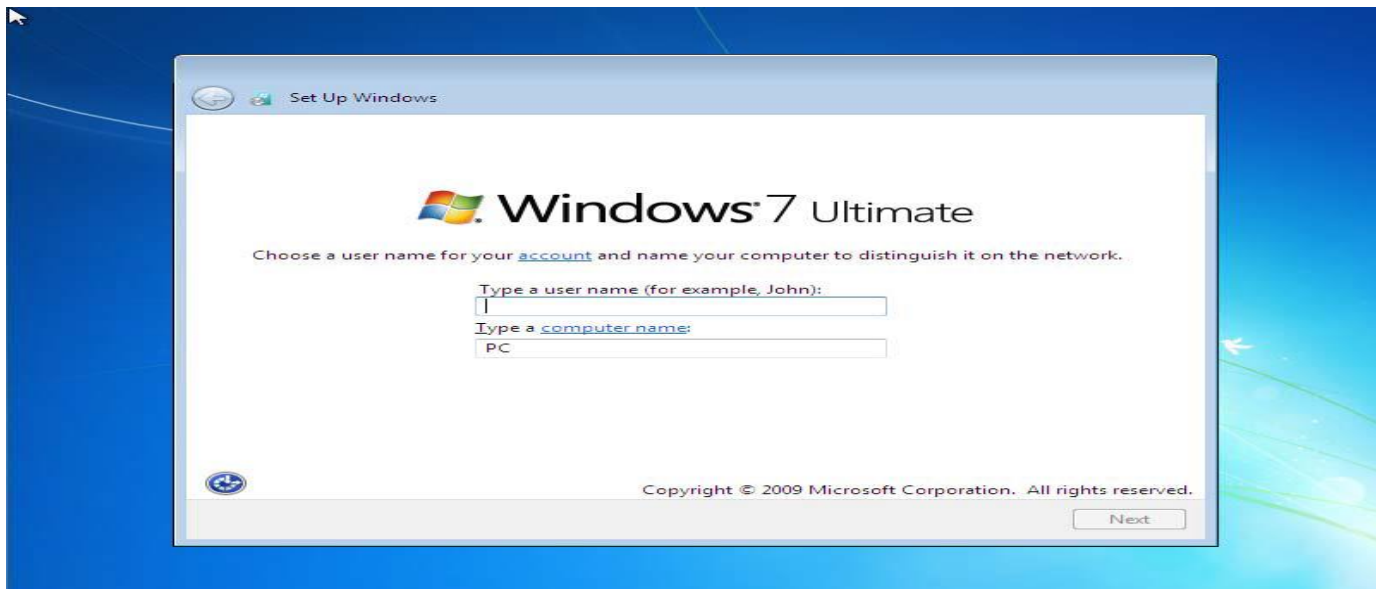
Step 10. Disk 0 Unallocated Space should be the only listing at this point. If it is press next, If not please proceed to delete any additional partitions that may be listed.



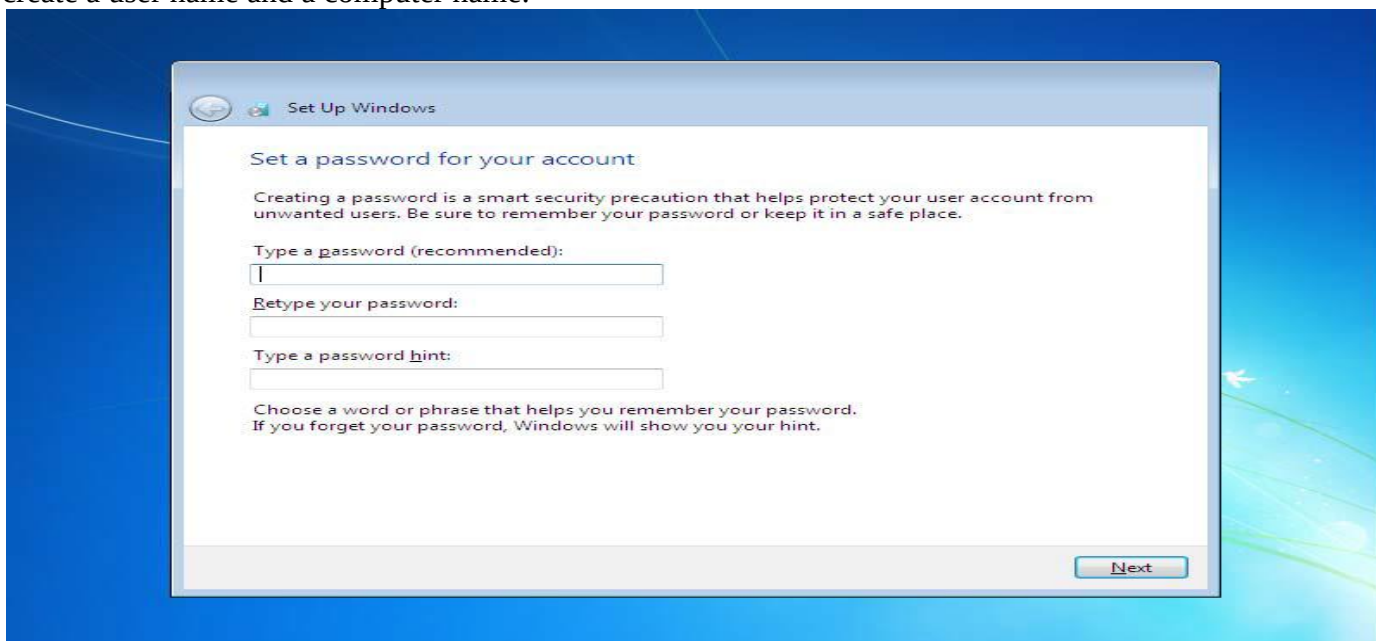
Step 11. The next screen will show “Installing Windows”. This process should take approximately 10 minutes after which the system will reboot

***NOTE:** Do not press any keys during this boot up process so as to not disturb the rest of the installation.

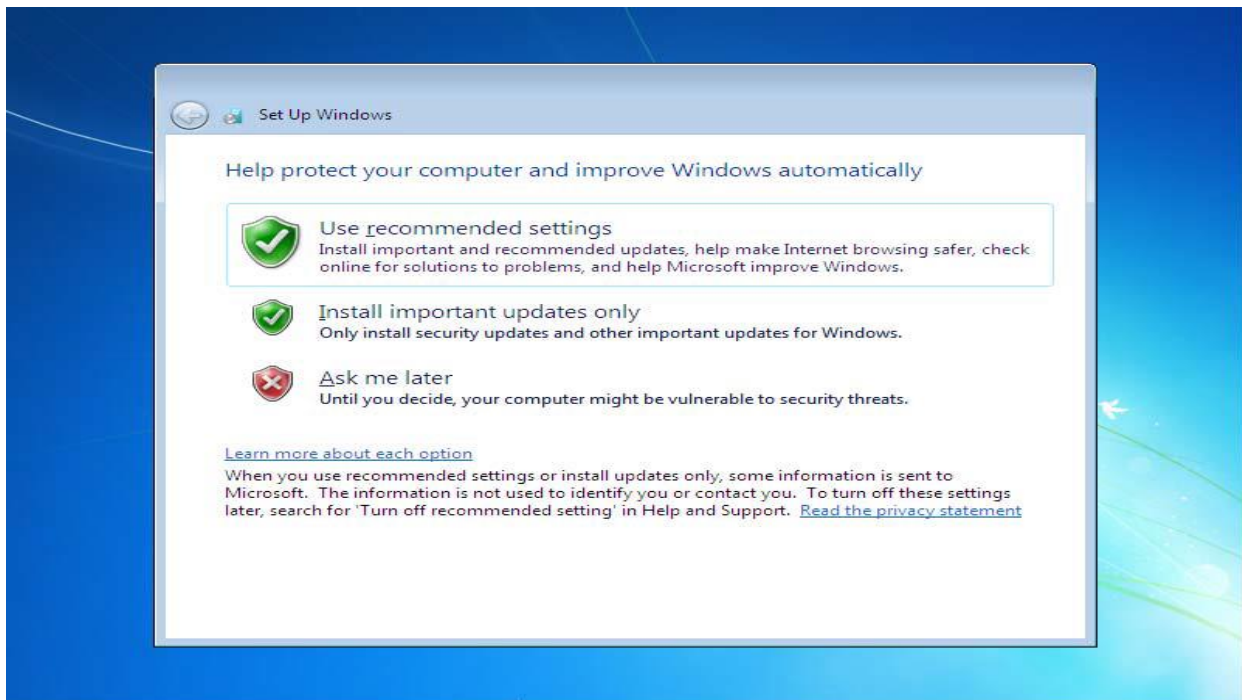
Step 12. The next screen will show “Installing Windows” again to complete the installation process.



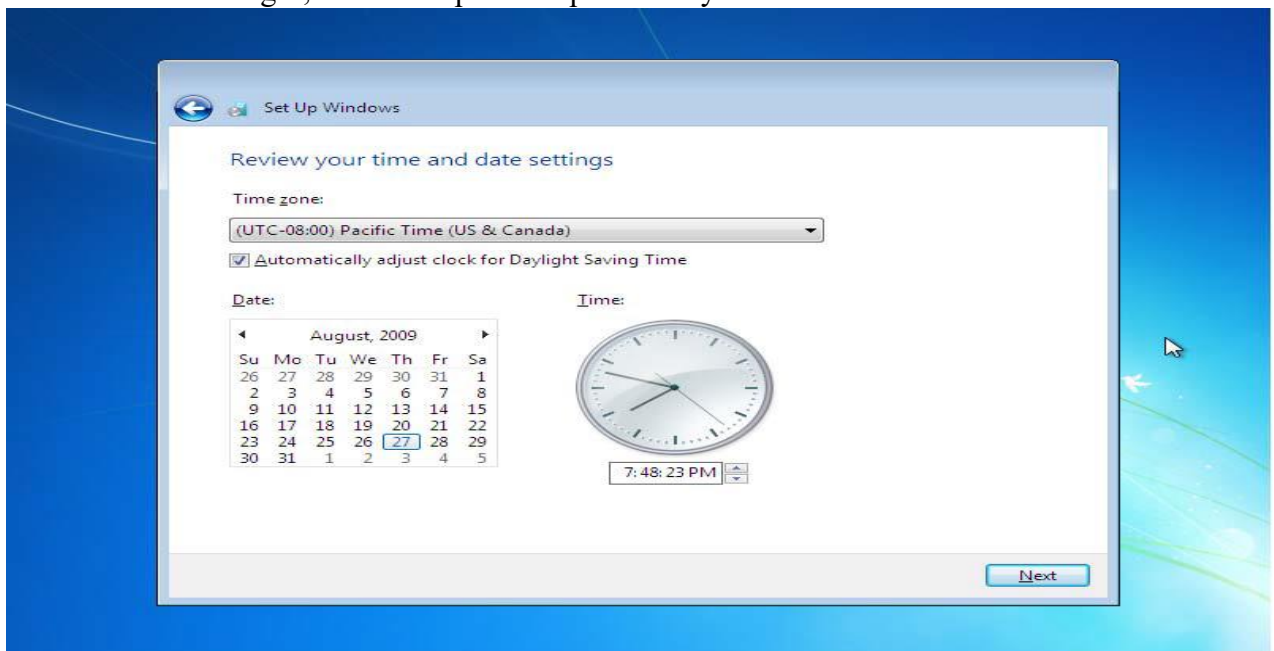
Step 13. “Setup is starting Windows” will appear on the screen, then you will be asked to create a user name and a computer name.



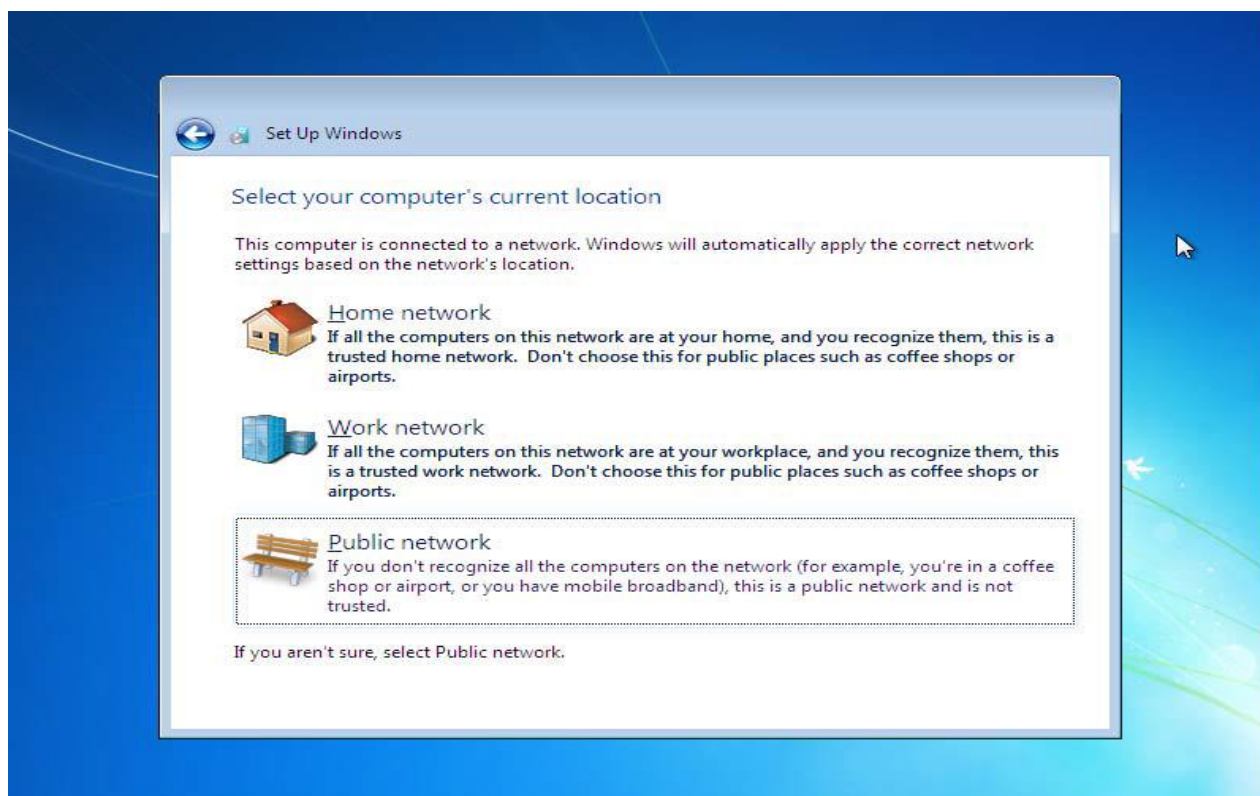
Step 14. The next step will prompt you to create a password for your account (optional).



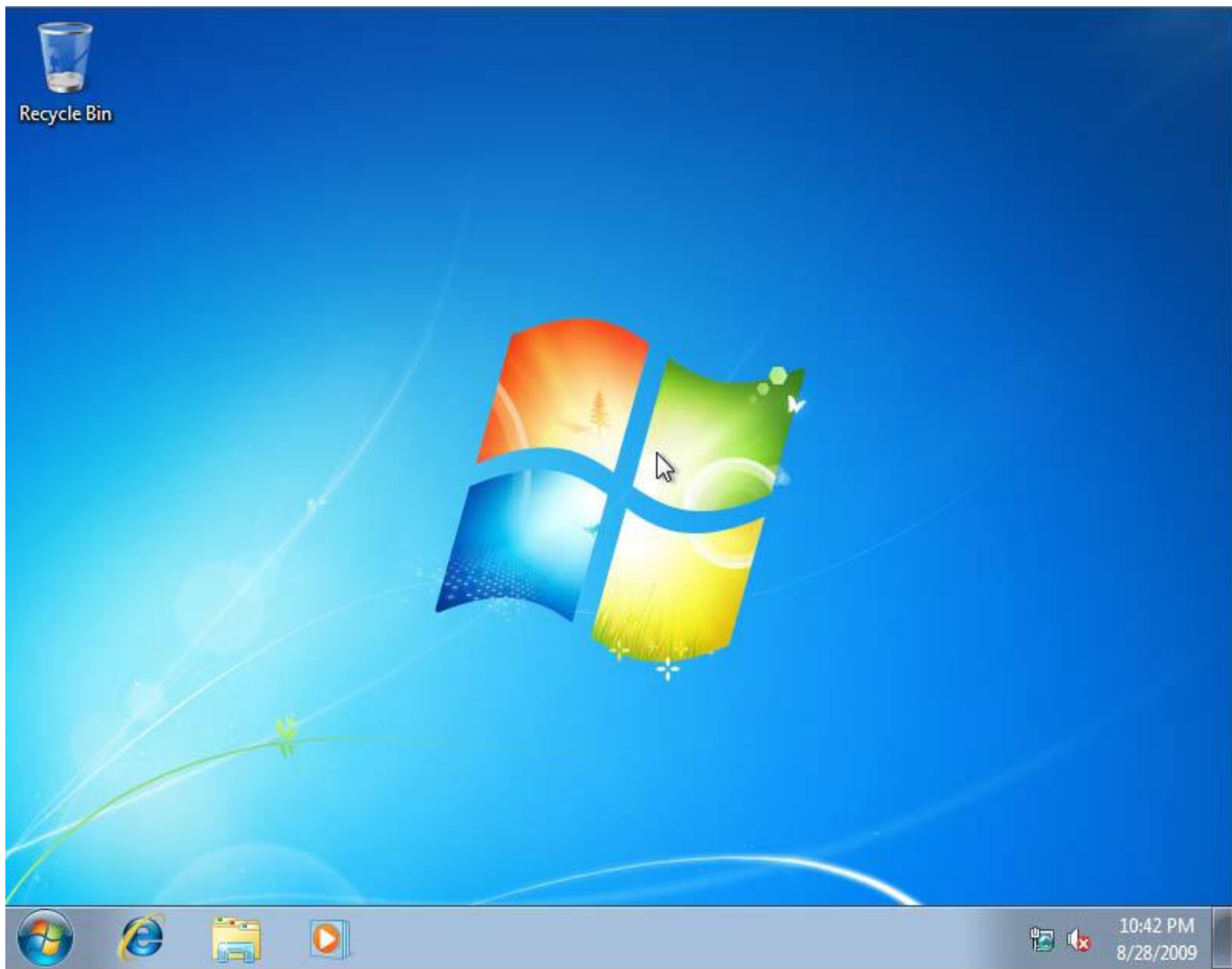
Step 15. The next screen will come up and ask you to choose one of three options: “Use recommended settings”, “Install important updates only” and “Ask me later”.



Step 16. This screen will allow you to choose your local time zone and also adjust the date and time.



Step 17. Click on the appropriate location of your computer to enable Windows 7 to apply the correct network settings.



The installation is done! You have successfully installed Windows 7 on your ZT computer.